

Empowering SAARC's energy future: A PESTEL-SWOT roadmap for super smart grids and P2P energy trading

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ABSTRACT

The super smart grid (SSG) is a revolutionary grid which offers significant fossil fuel elimination, emissions reduction, renewable energy integration, and demand fulfillment. However, such mega grids are in the strategic analysis stage due to the involvement of multiple countries and complexities. Although the existing literature has performed different types of analysis for the different SSGs around the world, there is a lack of studies on the strategic analysis of the SSG planned by the South Asian Association for Regional Cooperation (SAARC). For the first time, this review paper presents the hybrid PESTEL-SWOT analysis for the futuristic SAARC SSG. This paper offers important insights and strategies for the implementation of the futuristic SAARC SSG. For instance, a practical strategy towards the emergence of the SAARC SSG is the encouragement of the P2P trading at a very basic level through the hierarchical integration of thousands of prosumers, prosumer communities, and national grids.

Abbreviations

AI	Artificial Intelligence
HDI	Human Development Index
HVAC	High Voltage Alternating Current
HVDC	High-voltage Direct Current
IoT	Internet of Things
LCC	Line-Commutated Converters
P2P	Peer-to-Peer
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
PV	Photovoltaics
SAARC	South Asian Association for Regional Cooperation
SAFTA	South Asian Free Trade Area
VSC	Voltage Sourced Converter

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1. Introduction

This section presents the definition of the SSG, the objectives and novelty of this work, the major policy-level problem, and the need for the hybrid PESTEL-SWOT analysis. In addition, this section reviews the possibility of the SAARC SSG development, energy problems in the SAARC region, and the energy status of the SAARC region.

1.1. Super smart grids (SSGs)

Smart grids are more robust and eco-friendly than conventional grids. Conventional grids depend on fossil fuel-based power generation, resulting in substantial emissions into the natural environment [1]. Smart grids facilitate the integration of substantial renewable energy resources through different ways, such as prosumers, microgrids, and SSGs. The development of the SSGs is an emerging strategy in the world which would share the renewable energy-based electricity amongst different locations or countries across the world. This strategy will be beneficial for the South Asian region, where 25 % of the world's population lives in a relatively small area [2].

1.2. Objectives and novelty of this work

The key objective of this paper is to perform the hybrid PESTEL-SWOT analysis of the futuristic SAARC SSG. Table 1 highlights the contribution of this paper with respect to the existing literature. In the existing literature, most studies focused on the limited SSG factors, including cost estimation and feasibility assessment. However, the SAARC SSG can be affected by a range of other factors which can be addressed by the hybrid PESTEL and SWOT frameworks. In addition, the existing literature did not use the hybrid PESTEL-SWOT frameworks for the futuristic SSGs, especially the SAARC SSG. First, this paper uses PESTEL analysis to study the macro-environmental/external factors that would influence the success of the SAARC SSG. Then, the SWOT analysis was performed using inputs from the PESTEL analysis. In this view, the PESTEL analysis focuses on the external factors, and it can be used in the precise identification of the external factors (i.e. opportunities and threats) in the SWOT analysis. So far, no research has reported the integration of PESTEL analysis and SWOT analysis for the development of the SAARC SSG.

The development of the SSGs is a complex strategic decision due to the power generation, transmission, and distribution at multiple locations and multiple countries. This mega grid would require substantial funds from internal and external sources [11]. Therefore, any mistake in the strategic planning would result in unbearable losses to the stakeholders. Subsequently, many SSGs are in the strategic analysis stage. In this regard, most efforts are focused on the cost-benefit analysis. For instance, Reference [3] concluded that the North American SSG would result in substantial cost savings with wind power and solar power. Similarly, Reference [4] reported significant cost savings with the SSG between Central America and South America using 100 % renewable power. Likewise, the Chinese super grid would result in a reduction of 70 % emissions [5]. Another SSG has been proposed for Sub-Saharan Africa using 100 % renewable power. It has been claimed that the Sub-Saharan African SSG is technically possible and can satisfy the whole electricity demand of the region [6]. Cost-effectiveness and feasibility of several other SSGs have been studied for different countries or regions [12,13]. Similarly, Reference [7] investigated the potential of renewable energy utilization in the Indonesian super grid and concluded that the proposed super grid would achieve 100 % renewable energy with long-term investments in renewable energy and storage technologies. Reference [14] explored the potential feasibility of the SSGs and identified that the microgrids would be the most important building blocks for the futuristic SSGs. Consequently, extensive literature is available on the different aspects of SSGs around the world [15]. Despite the availability of significant literature, the implementation of SSGs is an emerging research area due to very slow progress in the adoption of this wonderful idea. Amongst the many potential SSGs, the SAARC SSG has been proposed for the South Asian region [16]. It has been identified that SAARC SSG would reduce high costs, decrease emissions, and satisfy electricity demand [17]. The key focus of the present paper is the strategic analysis of the futuristic SAARC SSG, which was planned by the SAARC organization for emissions reduction through the large-scale integration of renewable energy resources into electricity generation.

1.3. Major policy-level problem

The major policy-level problem in the development of the SAARC SSG is the absence of harmonized energy policies and regulations.

Table 1
Contribution of this paper with respect to the existing literature.

References	Focus area	Methods
[3]	Cost analysis for North American SSG	Scenario analysis using linear optimization of cost
[4]	Cost analysis for SSG between Central America and South America	
[5]	A Chinese super grid would result in a reduction of 70 % emissions	Scenario analysis using a simulation model PESTEL/SWOT analysis
[6]	Technical feasibility of Sub-Saharan African SSG	
[7]	Feasibility of Indonesian SSG	
[8]	Development of renewable energy in Togo country	
[9]	Identification of the key factors affecting the power system progress	
[10]	Evaluation of ocean and land-based floating solar projects	
This paper	Strategic analysis of SAARC SSG	

Additionally, there are inconsistent domestic policies for the improvement of the financial impact and infrastructure. National policies in the SAARC countries are mainly inclined towards fossil fuels. In addition, the cooperation between the SAARC countries is restricted. Moreover, there are inconsistent and high taxes on cross-border trade. Consequently, a range of internal and external factors would affect the development of the SAARC SSG. For the analysis of such factors, the PESTEL analysis and the SWOT analysis are the most commonly used tools. Consequently, this study uses SWOT analysis and PESTEL analysis to offer a strategic roadmap for the development of the SAARC SSG.

1.4. Need for the hybrid PESTEL-SWOT analysis

The development of the SAARC SSG is a futuristic plan in the SAARC region, just like other various possible SSGs around the world. Any futuristic plan requires a proper strategic analysis process to understand different dimensions and practicality of the proposed plan [18]. Hence, strategic analysis of the proposed SAARC SSG will be an appropriate and timely investigation. In the strategic analysis, PESTEL (political, economic, social, technological, environmental, legal) analysis and SWOT (strengths, weaknesses, opportunities, threats) analysis are the powerful tools for the analysis of external and internal environments of a strategic plan. The PESTEL analysis is a macro-environmental analysis technique, which evaluates the external factors influencing the success of a strategic plan [19]. SWOT analysis is used for the analysis of internal factors (i.e. strengths and weaknesses) and external factors (i.e. opportunities and threats) [20]. Regarding these methods, Reference [8] used the PESTEL and the SWOT methods to study the development of renewable energy in the Togo country. They concluded that the renewable energy utilization is gradually increasing in Togo, but there is still a need for greater investments and government support for the proper adoption of renewable energy technologies. Similarly, Reference [9] used the PESTEL and the SWOT analysis for the identification of the key factors affecting the power system progress in Togo. They highlighted that the power systems can be transformed through the advancement of electricity infrastructure and regional integration. Reference [10] performed the SWOT and the PESTEL methods for the evaluation of ocean and land-based floating solar projects. They explored the advantages and disadvantages of both systems. The PESTEL analysis evaluates the macro-environmental factors which are essential for highly complicated projects like SSG. Moreover, PESTEL analysis complements the opportunities and threats in the SWOT analysis. The SWOT analysis offers the analysis of internal strengths and weaknesses of the potential SSG and external opportunities and threats. The combination of both tools would offer robust and inclusive strategies for the analysis of the complications and challenges across the SAARC SSG. The hybrid PESTEL-SWOT offers an inclusive approach towards the analysis of external and internal factors affecting a mega project spanning across many countries.

1.5. Current status of the SAARC SSG possibility

The progress of the futuristic SAARC SSG is slow due to the various disputes between the SAARC countries [21]. The SAARC countries have many political, economic, and environmental challenges [22]. The SAARC team is working to address such challenges, but the SAARC team has limited actual support from these countries [23]. The SAARC team is making various efforts in various directions [24]. The SAARC countries are trying to integrate their infrastructure, commerce, and economic advantages [25]. Unfortunately, SAARC countries are lacking cooperation on sustainable energy sharing [26]. Hence, the regional energy sustainability is not the top priority of the SAARC countries [27]. No individual SAARC nation can address the climate change problems in an effective way [28]. On the other hand, the developed world has been cooperating in every possible way [29]. Through the appropriate collaboration and strategies, the SAARC nations would expedite the SAARC SSG [30]. Consequently, the strategic planning and analysis is the key step towards the successful and practical implementation of the SAARC SSG.

1.6. Energy problems in the SAARC region

In the SAARC region, the key electrical energy resources include natural gas, oil, coal, and hydropower [31]. These sources have detrimental impacts on human health [32] due to enormous CO₂ emissions [33]. In addition, the electricity demand and cost are continuously increasing in the SAARC region [34]. India is the largest energy consumer in the SAARC region [35]. The next largest energy consumers are Bangladesh and Pakistan. Nepal and Bhutan can provide hydropower to these countries [36]. The 100 % energy sharing between SAARC members should be based on renewable energy resources; otherwise, the SSG would result in more emissions [37]. However, the SAARC nations import a significant amount of fossil fuels from other nations, such as the Maldives [38].

To tackle the growing electricity demand, the construction of new power plants is infeasible [39]. A cost-effective way is the power sharing between the SAARC countries using high voltage transmission lines [40]. The SAARC region has a great solar energy potential in the world [41]. Hence, the greater solar potential would be the fundamental source for the SAARC mega grid [42]. However, there will be unique energy management challenges in the power system [43]. For instance, the solar power forecasting in the SAARC SSG is a complex and challenging task [44]. In addition, the wind power potential is also promising in the SAARC region [45]. However, the SAARC region is facing various power system issues such as load shedding, intermittent power outages, and the emission of dangerous gases [46]. To resolve these issues, the SAARC SSG offers an eco-friendly way to integrate renewable energy [47,48]. On the other hand, the European SSG would face fewer issues compared with the SAARC SSG [49]. In this perspective, Europe has set more strict targets towards the integration of renewable energy resources into the grids [50–53]. The SAARC SSG would offer significant cost savings [54], emissions reduction [55], and renewable energy integration [56]. In this regard, the SAARC team has already planned a mega power grid between the SAARC countries [57].

1.7. Energy status of the SAARC region

The renewable energy potential of the SAARC region is enough to construct a mega grid between the SAARC countries [58]. The key renewable energy resources in the SAARC region include solar power, wind power, hydropower, and biomass power [59–61]. Table S1 of the supplementary material presents the renewable energy potential for the SAARC countries. Table S2 of the supplementary material summarizes the peak power demand, peak generation, and installed capacity in the SAARC countries. It can be observed that the renewable energy potential of the SAARC region is immense and is enough to fulfill the demand of the whole region [62]. However, there are many other factors which need improvement to achieve the full advantage of this potential. This paper focuses on the strategic analysis of the important factors. In the following, the key issues in individual countries are summarized.

1.7.1. Afghanistan

Power system infrastructure in Afghanistan is poor due to economic issues and continuous wars. Afghanistan is suffering from a severe energy shortage and power system interruptions. Four operational networks are not completely connected to the national grid. There are major cities like Kabul and Kandahar that use voltage levels of 110 kV and 220 kV for electricity delivery [63]. A project for a 500 kV transmission line to import electricity from Tajikistan has been started in northern Afghanistan by the Asian Development Bank. The eight distribution feeders provide an average of 130 kWh of electricity per person annually at 11 kV. The country's queue losses are 25 % [64]. The best strategy for Afghanistan is to collaborate in the power sharing with other countries.

1.7.2. Bangladesh

Bangladesh is facing difficulties in providing power to the rural population. Moreover, electrified regions frequently experience inadequate power quality and fluctuating voltage levels [65]. Bangladesh cannot fulfill its electricity demand without fossil fuels [66, 67]. Bangladesh is unable to install solar, wind, and hydro power plants due to land limitations and unsuitability in the country [68–71]. The 400 kV, 220 kV, and 132 kV transmission lines that make up Bangladesh's extensive power network span 11,975 km in total. The distribution network has 33 kV lines. In addition, the line losses are 2.75 % [72]. Bangladesh is already importing power from Indian to fulfill its demand at shortage times.

1.7.3. Bhutan

Bhutan's people fulfill almost all electricity demand from the national grid [73], using hydro power [74]. In the season of water shortage, it uses imported fossil fuels and exports excess power to India in the time of excess water availability [75]. The solar PV/wind power installation is very low in Bhutan [76,77]. As a result, Bhutan is heavily dependent on hydro power [78]. However, water is gradually depleting in Bhutan, which would motivate Bhutan to participate in the SAARC SSG development. Already, Bhutan generates excess electricity that is transferred across a distance of 1124 km through 400 kV, 220 kV, and 132 kV lines. However, the transmission and distribution system is inefficient and vulnerable to voltage swings because of inadequate infrastructure and careless maintenance. In Bhutan, 1.25 % line losses have been recorded [79].

1.7.4. India

India is the largest consumer of fossil fuels in the SAARC region [80]. On the other hand, India has a greater potential for renewable energy resources [81]. Still, India has a poor electricity infrastructure, which results in grid outages for many hours [82]. India has an excellent opportunity to shift towards a renewable energy-based mega grid [83,84]. Before moving towards the SAARC SSG, India should establish its internal generation, transmission, and distribution networks. India's transmission system contains 400 kV AC, 765 kV HVDC, and 17,000 km of transmission lines. Also, the distribution network contains 11 kV and 33 kV lines. Additionally, the line losses are around 20 % [85].

1.7.5. Maldives

The Maldives contains various islands and a limited scope for the construction of a unified national grid [86]. Its whole electricity production is based on imported fossil fuels [87]. This country has a limited scope for the utilization of renewable energy resources [88, 89]. The geographical dispersion of the Maldives has caused a lack of a well-designed electrical transmission and distribution network. Line losses on the nation's 11 kV transmission and distribution lines are estimated to be 20 % [90].

1.7.6. Nepal

Nepal's national grid is unable to fulfill the whole electricity demand [91]. Nearly 42 % electricity demand of Nepal is fulfilled through renewable energy resources and imported electricity from India [92,93]. Nepal has a good potential for renewable energy-based power generation, but it is making negligible progress towards the renewable energy-based installations [94–96]. Hydro power is the major energy resource for Nepal. At the time of water shortage, Nepal faces severe load shedding issues [97]. The transmission and distribution infrastructure of Nepal is well established. The nation runs 11 kV distribution feeders in addition to 220 kV and 132 kV transmission lines. The nation's queue losses are approximately 4.64 % [98].

1.7.7. Pakistan

Major electricity generation resources in Pakistan contain fossil fuels [99]. Hydro power is also a significant electricity generation resource [100]. Despite enough capacity, the electricity cannot be reached in many locations due to financial and infrastructure issues [100]. Solar PV is emerging in Pakistan, but its contribution is very low [101]. The recent government policies do not encourage PV

installation [102]. In addition, there is a lack of funds for the renewable energy integration into the national grid [103–105]. The country possesses high-tension transmission lines with voltages of 500 kV, 220 kV, and 132 kV that span over 50,000 km over the whole nation [106]. Nevertheless, the system has many transmission and distribution losses, which contribute to power shortages and poor power quality. According to Reference [107], there are line losses of 20.2 %, which reduces electricity transmission to customers by over 20 GW.

1.7.8. Sri Lanka

Sri Lanka imports significant fossil fuels for electricity generation [108]. It is facing electricity shortage issues. Sri Lanka has a good potential for renewable energy, but it has installed a significantly low amount of renewable energy-based power capacity [109–114]. Sri Lanka's electricity distribution and transmission system is set up with 11 kV distribution circuits and 220 kV and 132 kV transmission lines. The Sri Lankan power system's transmission and distribution losses have been measured at 10.2 % [115].

The remainder of the paper is organized as follows. Section 2 presents the methodology used for the strategic analysis of the SAARC SSG. Section 3 performs the PESTEL analysis of the SAARC SSG. Section 4 presents the SWOT analysis based on the PESTEL analysis for the development of the SAARC SSG. Section 5 discusses a practical roadmap which would help in the journey towards the futuristic SAARC SSG. Section 6 concludes the paper.

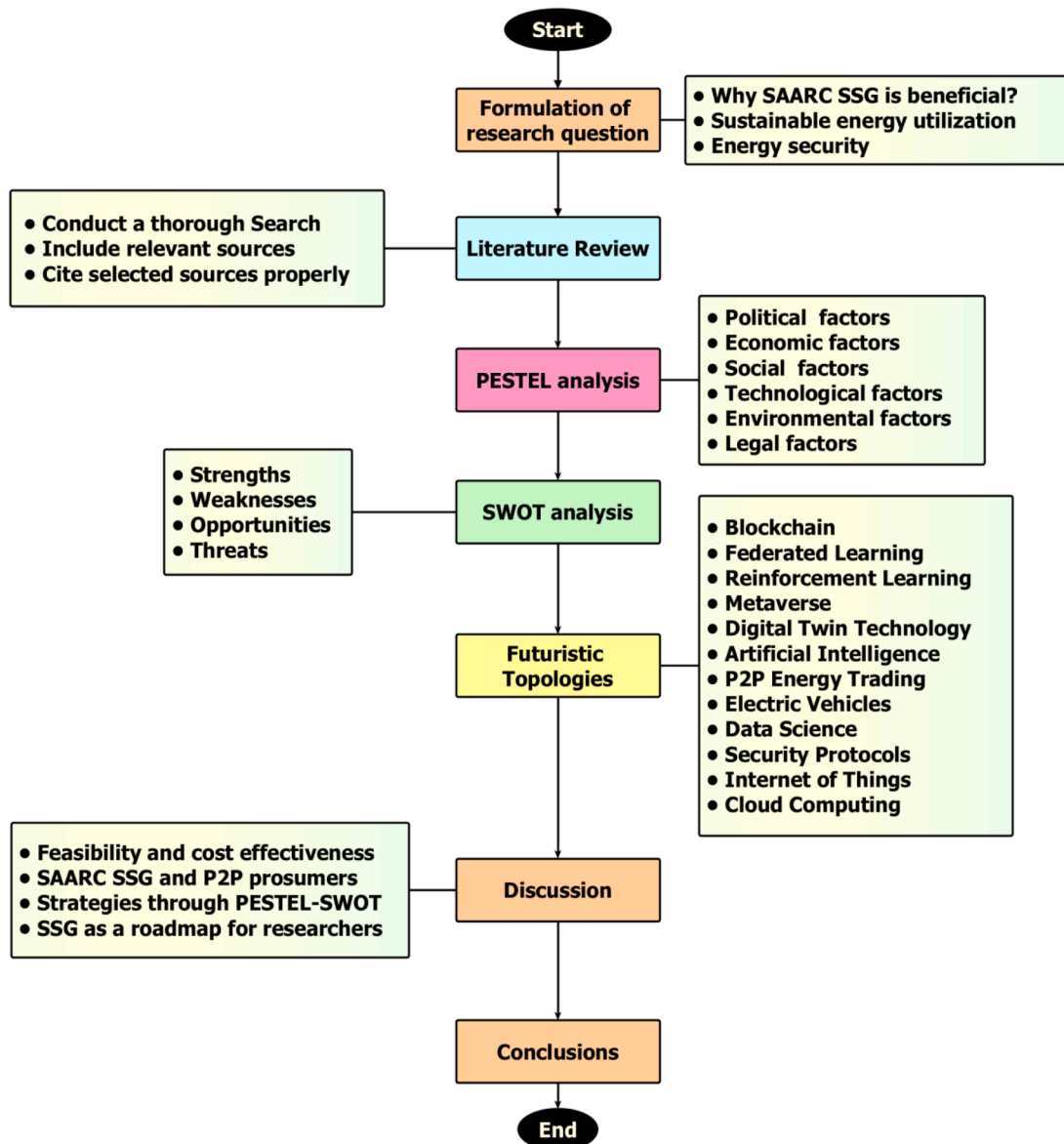


Fig. 1. The proposed framework.

2. Methodology

This study performs a PESTEL-SWOT analysis of the potential SAARC SSG based on the secondary data sources. To achieve this objective, this paper reviews a large amount of relevant data for the strategic analysis framework and devises practical strategies which would result in the emergence of a mega power grid between the SAARC countries. Fig. 1 presents the overall framework of the paper. Fig. 2 presents the PESTEL-SWOT framework methodology for the strategic analysis of the SAARC SSG.

The hybrid PESTEL and SWOT framework is a systematic integration of external factors from the PESTEL analysis into the SWOT analysis. In the first step, the external environment was scanned using the secondary data from the comprehensive literature review. Based on the external environmental scanning, the sub-factors were identified. In the second step, the internal environment was scanned using the secondary data from the comprehensive literature review. Based on the internal environmental scanning, the sub-

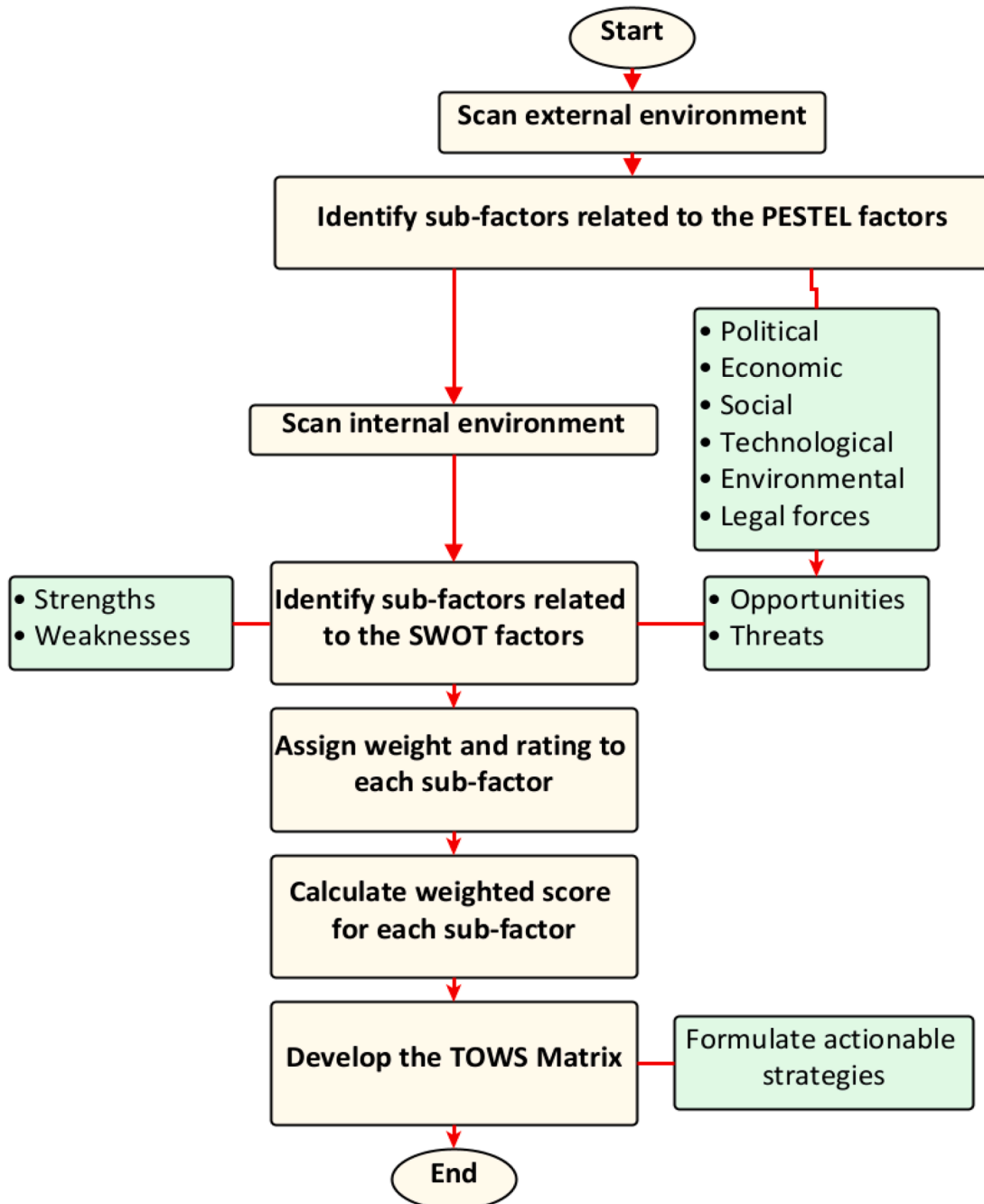


Fig. 2. PESTEL-SWOT framework.

factors were identified. In SWOT analysis, the opportunities and threats are the external factors. We included the positive factors from the PESTEL analysis as the opportunities, and the negative factors from the PESTEL analysis as the threats. Hence, the external factors from PESTEL analysis were used as input to the SWOT matrix.

In the SWOT matrix, the weight is the impact or importance of a factor. The sum of all weights should be 1 within a category (i.e. strengths, weaknesses, opportunities, threats). Similarly, the rating is the likelihood of a certain factor. We choose a scale of 0 to 1 for the rating, where 0 means the lower possibility, and 1 is the higher possibility for a factor. The weighted score was achieved by multiplying the weight and the likelihood of a factor. A factor with higher weighted scores for the strengths and opportunities shows that a factor is performing well. In this case, we should keep following the good performance of this factor. On the other hand, a factor with higher weighted scores for the weaknesses and threats shows that a factor is performing worse, and we should focus on this factor in order to prevent losses to our SSGs project. Finally, the TOWS Matrix was developed through matching the factors in the SWOT matrix. The TOWS matrix helps in the development of the actionable strategies.

3. PESTEL analysis of the SAARC SSG

For strategic analysis, the fundamental step is the macro-environmental analysis, which is the study of external factors influencing the success or failure of a project. The PESTEL analysis is the mostly used technique for macro-environmental analysis of a strategic plan [116]. Hence, this study selected this technique to understand the external factors, especially opportunities and threats, for subsequent SWOT analysis. Fig. 3 presents the six dimensions of the PESTEL analysis framework. In the following, these six dimensions have been evaluated for the development of the SAARC SSG. These dimensions offer comprehensive criteria for the possible implementation of the SAARC SSG [117]. The outputs of the PESTEL analysis are embedded into the SWOT framework to get realistic insights for the development of the SAARC SSG [118].

3.1. Political dimension

People and politicians of South Asian countries do not support the SAARC's efforts [119]. Initially, small SAARC countries were willing to support the SAARC objectives. However, negligible economic benefits and interstate rivalry discouraged them from maintaining this cooperation [120]. Despite multiple attempts by the SAARC, the political issues between the SAARC members have not been resolved. India has bilateral conflicts with Pakistan, Bangladesh, and Sri Lanka [121]. As a result, these countries did not give a positive response. India, Pakistan, Nepal, Bangladesh, and Sri Lanka are engaged in diverse ethnic, border, and water disputes. These conflicts resulted in the cancellation of 50 % planned SAARC summits [122]. In the annual summits, political leaders of the region pledge to enhance mutual trade and cooperation between SAARC countries. The South Asian Free Trade Area (SAFTA) agreement is a major agreement between SAARC countries, but various internal and external political issues prevent its implementation.

The internal political systems of the SAARC members are facing major challenges. Fig. 4 presents the democracy ranking of the SAARC countries based on political freedom, political equality, and political and legal control (www.democracymatrix.com). Total Value Index is based on a 0–1 scale, where 1 represents maximum democracy. It can be observed that the SAARC countries have a lower democracy index, and there is a difference in their political systems. Challenges to democracy discourage the involvement of

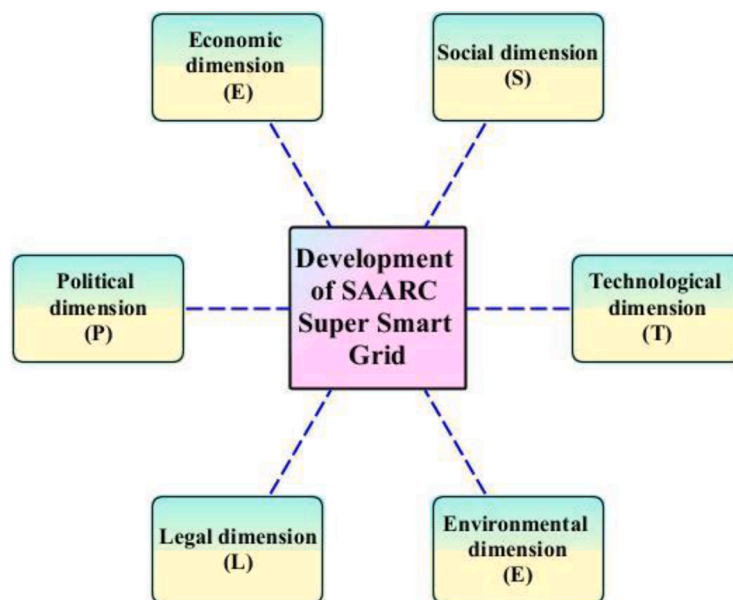


Fig. 3. PESTEL analysis for SAARC SSG development.

people in different matters. The ranking of the democracy has a direct impact on the SSG's success. The highly democratized system is more effective with less corruption and more participation of the people. For instance, the allocation of funds will be more transparent. Also, the involvement of people will result in the strategic success of the SSG implementation. Democratic nations often decide based on the thorough knowledge from multiple sources, which is necessary for the SSGs, which involves the majority of people at the prosumer level.

Policies, laws, and regulations of all the SAARC countries are different and inconsistent. Corruption is another issue which would affect the mutual trade between the SAARC members. It can be concluded that the above political issues would hinder the development and maintenance of a mega power grid between SAARC members. Table 2 shows the political barriers to the SAARC energy cooperation and the SSG development. SAARC's annual summits usually result in the signing of agreements for boosting bilateral trade and cooperation, but the local political instability makes these efforts useless [123]. Political systems of the SAARC countries are very diverse, and many have governance issues such as fragile democratic institutions, corruption and inconsistent regulatory practices [124]. The low position held by the SAARC countries in international democracy rankings shows deficiencies emerged from governance and a weak public presence in policy-making [125]. These governance dilemmas prevent regional cooperation and investments, specifically in the case of major infrastructure. A high level of political commitment, agreement on policies and stable governance structures are required for the robust energy management and successful regional interconnection [126]. However, there are significant barriers due to the discrepancy in energy policies and regulatory frameworks in the SAARC member states. For example, the internal corruption and bureaucratic inefficiencies in the individual SAARC members have resulted in delays to cross-border energy projects due to misalignment of policies [127].

Political mistrust between the SAARC states also obstructs the flow of information and decision-making on energy infrastructure development. The region is rich in renewable resources, but the techno-political conflicts prevent the effective energy trade and grid integration [128]. Lack of a cohesive regional energy strategy has resulted in fragmented and isolated national energy policies. Some SAARC member states have signed bilateral energy agreements; however, a regional power grid is difficult [129]. For instance, India and Pakistan are competing on a geopolitical level, which may hinder the efficient transmission across borders [130].

Political challenges in the SAARC, from bilateral disputes and governance failures to regulatory inconsistencies, constitute a serious impediment to regional cooperation on energy management and interconnection. The future of a sustainable and linked regional energy network comes into question unless these political challenges can be overcome. The SAARC countries need to focus on diplomacy, create common regulatory templates, and implement confidence-building measures so that successful regional energy integration can be achieved despite these impediments [132].

3.2. Economics

South Asian countries have a lack of funds for the construction of such a high-level infrastructure [131,132]. Economic factors such as higher inflation rates, interest rates, and unemployment rates are preventing the economic growth of this region, resulting in the

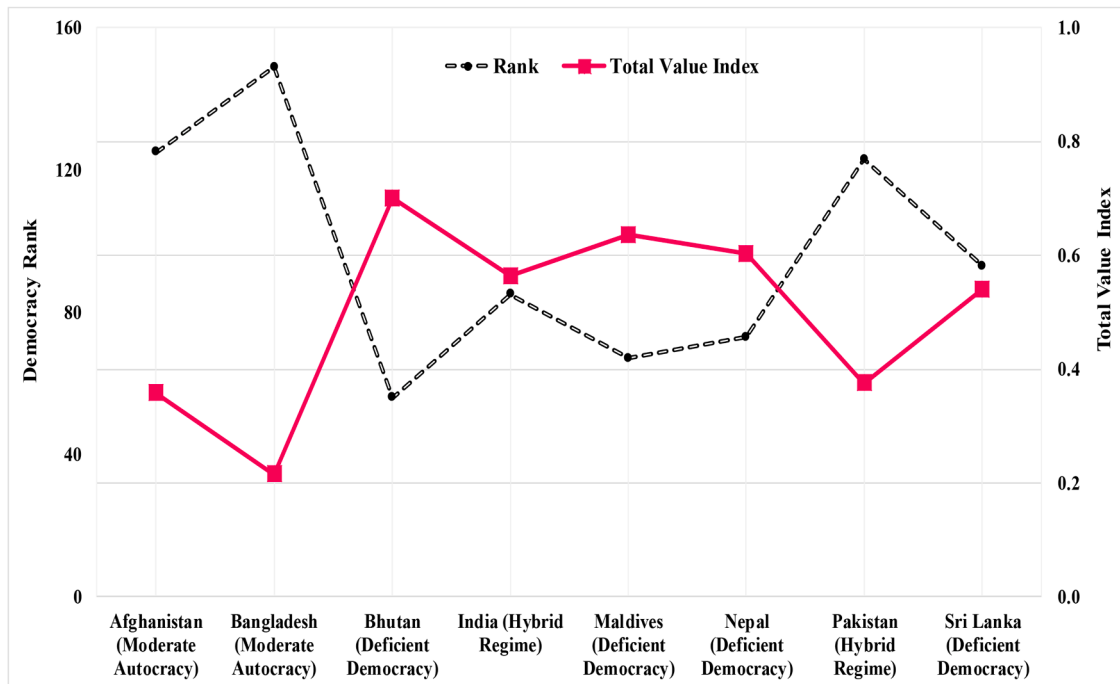


Fig. 4. Democracy ranking of the SAARC countries.

Table 2
Political barriers to SAARC energy cooperation and SSG development.

Challenges	Political Issues	Impact on Energy Management & Interconnection
Interstate Conflicts	Bilateral tensions among SAARC nations, particularly India-Pakistan, India-Bangladesh, and India-Sri Lanka.	Lack of trust and cooperation in energy projects; failure to establish joint energy agreements.
SAARC Summit Cancellations	50 % of summits were cancelled due to political disputes.	Limited diplomatic engagement; delays in regional energy initiatives.
Governance Issues	Weak democratic institutions, political instability, and corruption.	Inefficient policymaking and, lack of standardized energy regulations.
Trade Agreement Failures	SAFTA remains unimplemented due to political and economic challenges.	Hinders regional economic cooperation, limiting energy trade and investments.
Divergent Energy Policies	Lack of harmonized policies and regulatory frameworks among SAARC members.	Barriers to cross-border energy trade and power grid interconnection.
Lack of Diplomatic Trust	Absence of legally binding energy cooperation frameworks.	Prevents the establishment of a regional energy strategy and joint infrastructure projects.
Geopolitical Rivalries	India-Pakistan tensions prevent energy trade and grid connectivity.	No regional electricity-sharing mechanism; reliance on isolated national grids.
Underutilization of Renewable Resources	Political conflicts hinder energy trade between countries rich in hydro and solar power.	Inefficient use of renewable resources; no integrated energy infrastructure.

economic crisis [133,134]. In addition, the unavailability of external and internal funds for such a mega grid could prevent its construction. Despite, continuous increase in export taxes would increase the overall power generation and transmission cost. In this regard, there are unclear and inconsistent policies about taxes on imports and exports. Moreover, political issues are the major hurdles in the economic cooperation process.

The SAARC countries are hesitant to accept the dominance of India in the key economic areas. Similarly, India is slow in cooperation with other SAARC countries related to its domestic markets. India has the largest portions of people, infrastructure, and technologies, which results in the economic imbalance between SAARC countries [135]. For example, the GDP of India is approximately 77 % of the whole of South Asia, as shown in Fig. 5. Pakistan allows limited trade with India due to strategic concerns, such as the fear of market loss to Indian companies [136]. The SAARC countries are facing a funding shortage for various development projects. Often, funds allocated for development projects are shifted towards more emergencies. This situation would delay the development of the SAARC SSG. Lack of funds would cause hurdles in the technological implementation related to SSG. The weak domestic economic situation of each SAARC country is not favourable for regional integration. Instead of economic cooperation, each SAARC country is competing for integration with the developed world. Limited trade between the SAARC countries is preventing

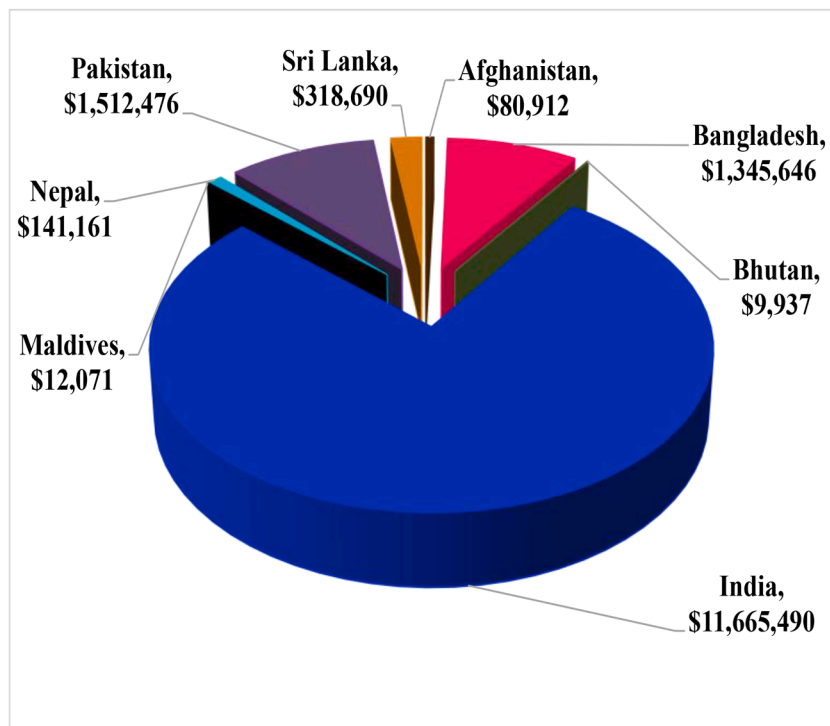


Fig. 5. GDP (PPP) (in millions).

economic growth. Seamless trade between South Asian countries would result in great economic gain. The SAFTA agreement is a wonderful achievement of SAARC for the free trade between SAARC members [137]. These countries are importing high amounts of expensive oil around the world. Alternatively, the SAARC SSG would result in the renewable energy between these countries. As a result, mutual economic benefits can be achieved. Sadly, some major SAARC countries have missed the excellent economic growth opportunities due to political unwillingness.

The severe economic realities faced by South Asia pose significant obstacles to the development and roll-out of high-tech infrastructure projects such as the planned SSG, which is intended to connect energy systems throughout the region for enhanced sharing [138]. However, the lack of funding serves as an obstacle to such an initiative. The growing inflation and high unemployment contribute to the general economic crisis that damages the long-term growth and development. These financial constraints prevent South Asia from reserving enough resources for building mega energy grids that require the immense up-front investments [139]. The lack of external and internal financing for large-scale projects transforms this into an even greater challenge that can prevent progress in the regional energy integration.

The inconsistent taxation policies increase the economic burden of the SAARC nations. For example, duties on export and import are unclear and inconsistent, which creates unpredictability in the energy generation and transmission costs [140]. Consequently, the public and private sectors are discouraged from investing in the energy projects. South Asian countries were relying on this expensive fossil fuel despite the severe increase in oil prices [141]. The South Asian region requires a transformational approach to organize the natural resource allocation and minimize energy [142]. The renewable energy sharing through the SAARC SSG would prevent the expensive oil import in the region. In addition, the overall cost would decrease and diverse sustainable energy sources would be added in the energy mix, resulting in the enhancement of the regional as well as national security [143]. The political conflicts in the SAARC region discourage economic cooperation [144]. In fact, several SAARC countries have limited trade relations due to competition and other conflicts [145]. The SAARC team has a limited authority despite opportunities for the regional trade agreements through energy interconnections. The SAARC members have an inability to agree and compromise on some important economic issues. Instead of competing for how they can work together in ways that would benefit their collective economic development, countries compete to be the most integrated with one or another more developed nation at a regional level [146].

For the successful SAARC SSG, the economic and political weaknesses should be overcome [147]. These obstacles could be mitigated to a significant extent by policies that enable investments in renewable energy infrastructure and provide financial incentives for regional collaboration. The proposed regional grid would offer economic stability, which would supersede the severity of other conflicts [148]. In this regard, there are significant economic problems in South Asia covering funding, inflation and policy inconsistencies, but those issues would be outweighed by a potential for beneficial regional energy interconnection with considerable economic advantages [149]. Table 3 presents the economic barriers to the SAARC energy cooperation and the SSG development.

3.3. Social

There is a lack of proper availability and implementation of health and safety rules in the SAARC countries, which would threaten the safety of workers involved in the running of such a large-scale grid. In addition, there are issues related to the implementation of labour laws. Another major issue is the differences in the cultural values between the SAARC members. The European Union has integrated their political and economic systems. In addition, European Union countries have many cultural similarities. In addition to political and economic issues, the SAARC countries have a range of cultural differences and clashes. These differences result in the lack of social integration with opponents. These countries did not facilitate their people for frequent social integration. For example, there

Table 3
Economic barriers to SAARC energy cooperation and SSG development.

Category	Economic Challenges and Causes	Impact on Energy Cooperation and SSG
Lack of Infrastructure Funding	- SAARC countries lack sufficient financial resources for large-scale infrastructure projects. - External and internal funding sources for the SAARC SSG remain unavailable.	- Delayed or abandoned energy interconnection projects. - Increased reliance on expensive energy imports.
Economic Instability	- High inflation, interest rates, and unemployment restrict economic growth. - Funds are often redirected to emergencies, delaying energy infrastructure development.	- Weak domestic economies make long-term investments in SSG unfeasible. - Increased financial uncertainty discourages private sector participation.
Trade & Taxation Issues	- Rising export taxes increase power generation and transmission costs. - Unclear and inconsistent tax policies on imports/exports create trade barriers.	- Higher energy costs discourage renewable energy investment. - Limited cross-border electricity trade due to unpredictable tariffs.
Geopolitical Tensions & Rivalries	- India's economic dominance. - Pakistan restricts trade with India due to fears of market loss. - SAARC countries compete for global economic integration rather than regional trade.	- Limited cooperation on energy projects due to geopolitical mistrust. - Lack of a unified regional energy market delays SAARC SSG progress.
Missed Economic Opportunities	- The SAFTA agreement promotes free trade, but political barriers hinder its full implementation. - SAARC countries import large amounts of expensive oil, rather than developing a shared renewable energy network.	- SAARC SSG could enable renewable energy sharing, reducing costs and reliance on fossil fuels. - Political unwillingness prevents mutual economic benefits from regional energy cooperation.

are limited facilities for air or road travel. However, South Asian countries have enough opportunities for social cooperation. For instance, the labour mobility between different SAARC countries would raise the social standards of people. SAARC has identified several areas for social integration of its member countries, which include health improvement, human resource development, poverty reduction, education, and skill development. Human rights violations have been reported in South Asian countries. The United Nations developed the human development index (HDI) to support the social development of the world. Fig. 6 presents the HDI and world ranking for SAARC countries (<https://hdr.undp.org>). The HDI offers a beneficial index of social progress and quality of life in the region [150]. The SAARC countries have low-to-medium development ratings as per the HDI rankings, and so this depicts that there is an urgent need to reduce social disparities if the SAARC members want the regional cooperation [151]. Strengthening the labour mobility, human resource development and protection of basic rights will enable the SAARC countries to improve their HDI rankings as well as promote equitable economic and social developments in the region.

There are several complicated issues to be tackled on the social front in the SAARC region, which if not planned and handled properly then executing a mega super grid project will only remain an academic solution [152]. We hear about the economic and political challenges to such a project, but various social reasons are critically important for the regional energy integration and management, whether labour mobility and cultural differences or health and safety regulations [153]. Currently, the SAARC countries still face some of the greatest social challenges. For a project like the SSG, the social challenges should be addressed [154]. The lack of universal health and safety measures in the SAARC countries poses a huge risk to workers. If workers are exposed to harsh conditions while building and maintaining the grid, not only would their lives be at risk, but an investment could be wasted [155]. Since many energy infrastructure projects in the region entail high-risk activities, including electrical work and transmission at extremely high voltage levels as well as installation of renewable energy technologies, it is important to establish strict safety protocols that can help prevent accidents while safeguarding human dignity [156].

Apart from health and safety, the implementation of labour laws is another area of concern. Different applications of labour laws in SAARC countries may fragment the workforce and create inequity between workers' rights [157]. Labour protection is often lacking in many countries of the region, and workers frequently work under poor conditions and receive low wages with little or no social benefits. Mega projects like the SSG require many different types of labour from heterogeneous countries to work across borders; this lack of standardization might lead to mistreatments, conflicts and delays [158]. Protection of labour rights, enforcement of labour standards, and a respectful work environment are needed for the construction of SSG.

Cultural diversity difference is another great social problem between the SAARC countries [159]. The SAARC region is characterized by very strong divergent cultures, such as language differences, religious beliefs, and traditions. These cultural differences potentially have implications for labour mobility and efforts at social integration between nations [160]. These differences in culture can require the solution to various issues such as communication difficulties, lack of teamwork, and cooperation [161].

Labour mobility and social integration are limited between the SAARC countries [162]. The mobility problem is limiting the movement of skilled labour, a necessity for many types of successful mega infrastructure projects, which require specialist workers. In addition, the lack of convenient transportation facilities would cause difficulties in the training or exchange of experiences for SSG

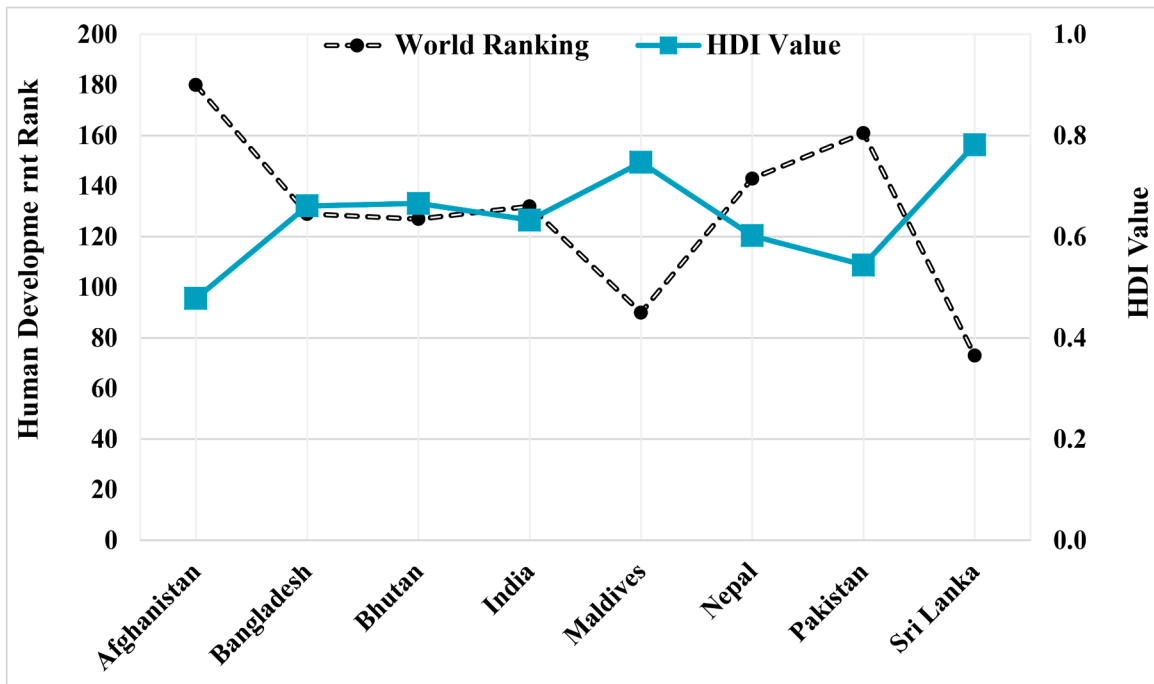


Fig. 6. Human development index (HDI) Ranking of SAARC countries.

[163]. Nevertheless, there are immense possibilities for social co-operation and development in the SAARC zone for energy sharing. The SAARC SSG would significantly enhance the social progress [166]. The result of the social issues would raise the quality of life for individual workers and regional togetherness [164]. Additionally, the SAARC has also identified priority fields for social integration, which can form an important input in the process of SSG implementation. These are areas such as health and safety, human resource building, poverty reduction, and education [168]. For example, a properly executed SSG could offer skill development for renewable-energy technology-based jobs in the future, potentially providing workers with high-demand technical capability that increases their job readiness and ability to signify an eco-friendly income [165].

Human rights are another crucial aspect of social integration that cannot be overlooked. Reports of human rights violations in some SAARC countries highlight the need for improved governance and protection of individual freedoms across the region [166]. A large-scale infrastructure project like the SSG could provide an opportunity to establish more robust human rights protections, especially for migrant workers who may face exploitation or abuse [167]. Ensuring the fair treatment of workers and upholding human rights would be essential not only for the success of the project but also for developing greater trust and cooperation between SAARC countries.

Social challenges in the many SAARC countries are immense. The mega projects like SSG can create large opportunities for energy interconnection and management as a way to deal with these same issues [168]. The region can better integrate a more sustainable workforce for large energy infrastructure projects by increasing labour mobility, improving health and safety laws, advancing cultural understanding, and enhancing social integration. The achievement of the SSG success can act as a driver to initiate greater social and economic development, which is beneficial for the energy sector and the common people in South Asia [169]. Table 4 shows social barriers for the SAARC region.

3.4. Technological

The SAARC members have a golden opportunity to adopt modern technologies in the SSG construction. The SAARC SSG can adopt these technologies in parallel with the evolution of other SSGs around the world. It has been reported that the SAARC region has adopted least technological innovations compared with other regions. In the SAARC countries, technology up-gradation is linked with the economic situation, which would cause issues in the SAARC SSG. Also, insufficient funds are allocated for research and development activities. The SAARC region has a weak infrastructure for technology upgradation. South Asian countries prefer technological and scientific collaborations outside the region. Fortunately, most of the SAARC countries have historical ties and language similarities, which would trigger the research and development on the SAARC SSG.

The most powerful technologies for SSG development are high voltage direct current (HVDC), high voltage alternating current (HVAC), Ultrahigh-voltage direct current (UHVDC), voltage-sourced converter (VSC), line-commutated converter (LCC), and flexible alternating current transmission system (FACTS). The applications of these technologies are well understood in the literature. HVDC,

Table 4
Social barriers to SAARC energy cooperation and SSG development.

Category	Social Challenges and Causes	Impact on Energy Cooperation and SSG
Health and Safety Issues	- Lack of proper health & safety regulations for workers in large infrastructure projects. - Exposed to hazardous conditions without adequate protection.	- Increased risk of accidents and fatalities. - Potential delays and cost overruns due to workforce injuries.
Enforcement of Labour Laws	- Poorly implemented or inadequately enforced labour laws across SAARC countries. - Labour exploitation and poor working conditions are prevalent in many projects.	- Lower worker morale and productivity - Inconsistent worker welfare across borders. - Labour unrest and strikes could delay or halt projects. - Difficulties in managing a mobile workforce across diverse labour standards.
Cultural and Social Integration	- Cultural differences (e.g., language, traditions, and religion) hinder cooperation. - Lack of shared values and trust between SAARC nations	- Complicated cross-border cooperation on social and economic projects. - Limited social integration prevents cohesive regional development. - Hindered joint decision-making and long-term project success.
Labour Mobility and Physical Connectivity	- Limited infrastructure for travel (air and road) within the region. - Poor transportation networks hinder worker movement.	- Difficulty in mobilizing skilled labour across borders for projects. - Lack of social cooperation and limited labour market opportunities. - Physical barriers to workforce training and development.
Human Rights Concerns	- Human rights violations are reported in various SAARC countries. - Unequal access to basic services, including education and health.	- A negative environment for social cooperation. - Inequality and discrimination within the region may affect worker morale. - Slower progress towards sustainable development goals.
Human Resource Development	- Weak human resource development programs within the SAARC nations. - Limited access to education and skill development for workers.	- Skills gap in the workforce for large-scale infrastructure projects like SSG. - Difficulty in training a skilled, mobile workforce for regional cooperation.

HVAC, and UHVDC can be used based on the requirements in the SAARC SSG. FACTS devices will be more beneficial to increase the stability of the SAARC SSG.

The robotic power plants are promising technologies for the SAARC SSG [170]. The various latest technologies would be highly desirable for building efficient, flexible and resilient electric transmission networks [171]. Although the technological transformation know-how is available in the literature but the SAARC technology adoption and innovation are far behind the rest of the world [172]. Technology up-gradation is directly proportional to the financial position in many SAARC countries [173]. The SAARC nations require a significant investment in research and development for the effective adoption of advanced grid technologies [174,179]. The technology infrastructure is weak in the SAARC region [175]. The common languages and culture in different parts of the SAARC region offer a conducive atmosphere [176–178].

The HVDC (High Voltage Direct Current) is a useful technology for long-distance, high-capacity transmission. This enables minimum transmission losses and facilitates better power exchange between countries having various grid frequencies. This is a technology that would be essential for the SAARC SSG. The region is geographically diverse, and so the ideal technology to transmit power would be the HVDC lines that are capable of transmitting large amounts of power across vast distances [179]. Similarly, the HVAC (High Voltage Alternating Current) can be used at various places in SSG [180]. UHVDC (Ultra-high voltage direct current) systems are built on the technical concept of HVDC systems by allowing power to be transmitted at higher voltage levels, thereby enabling even longer distances and large amounts of electricity to flow through those links [181]. VSC (Voltage-Sourced Converters) contributes to making power control in the power system more flexible, especially for connecting large-scale renewable energy and keeping a steady state when the supply changes trend is fast [182]. They are therefore most suitable for use in decentralized power systems, such as those utilizing solar or wind power, where the flow of electricity fluctuates. Superior voltage and power flow control can help stabilize the regional grid, which will be especially important in Germany when it decommissions its nuclear facilities as scheduled [183]. In the same way, LCC (Line-Commute Converters) combined with HVDC systems convert AC power into DC and vice versa. This is particularly beneficial for bridging the various grids of varying frequencies together to deliver a consistent and constant level of energy [184]. FACTS (Flexible AC Transmission Systems) devices are important for enhancing the stability, reliability, and efficiency SAARC SSG. FACTS can be used to improve voltage control, power flow management and alleviate the need for new transmission facilities, therefore assisting in integrating additional regional renewable energy resources into the grid. They are especially useful for grid resiliency in grids with significant amounts of renewable energy, as they can respond rapidly to rapid changes in power supply [185]. Several studies have been published regarding the wide range of applications for these technologies, and could contribute to a significantly higher efficiency and reliability of the SAARC SSG if they were used [186]. The changing global energy landscape gives the SAARC region time to implement these technologies in conjunction with the latest developments taking place elsewhere around the world, combining various best practices and adapting them as per local requirements [187]. The SAARC countries face hurdles to employing modern energy technologies, primarily because of economic and investment limitations, but there is an auspicious possibility for technological innovation through collaborative efforts [188]. The SAARC countries can develop an advanced, efficient and robust SSG that allows regional energy sharing across borders with ease; optimizes renewable energy penetration to the highest levels possible at the lowest cost of generation, which will contribute towards climate change mitigation [189]. The investment in technological adoption investment, will only be possible through more international partnerships in the region; teamwork across country borders is almost indispensable at this point. Table 5 shows technological barriers, and Table 6 presents the recommendations to the

Table 5
Technological barriers to the SAARC energy cooperation and SSG development.

Category	Challenges	Opportunities/Technologies
Technological Adoption	• Slow adoption of modern technologies due to economic constraints. • Underutilization of advanced technologies like HVDC and FACTS.	• HVDC and UHVDC for long-distance transmission of renewable energy. • HVAC for short-distance transmission and interconnecting grids. • VSC and FACTS for flexible grid control, especially with renewable energy.
Economic Constraints & Funding	• Limited funds for research and development (R&D). • Competing national priorities affect resource allocation.	• Prioritize cost-effective technologies (e.g., HVDC, FACTS). • Potential for regional R&D collaboration to develop sustainable solutions.
Weak Infrastructure for Up gradation	• Lack of research institutions and laboratories for local technological advancement. • Insufficient infrastructure for scaling advanced energy technologies.	• Establish regional innovation hubs to foster technological development. • Leverage historical and cultural ties to improve social and technological integration.
Integration of Renewable Energy	• High variability of renewable energy causes grid instability. • Existing grids are not designed for high penetration of renewable energy.	• VSC and FACTS stabilize the grid during fluctuating power supply. • Flexible AC transmission systems optimize solar and wind integration.
Regional Cooperation and Knowledge Sharing	• Lack of unified policies and standards among SAARC countries. • Political tensions hinder collaboration.	• Regional cooperation accelerates the adoption of advanced grid technologies. • Knowledge exchange platforms promote shared learning. • Cross-border partnerships with developed countries bring expertise and funding.

SAARC energy cooperation and SSG development [190].

3.5. Environment

Nearly 25 % of the world's population lives in the SAARC region. In this region, electricity demand is increasing as a result of rapid population growth and lifestyle upgrades. For major energy needs, this population is relying on fossil fuels, mainly imported oil, coal, and gas. The SSG is only useful if the member countries share only renewable energy-based electricity. The SAARC region has abundant renewable energy resources. The recent power demand of the SAARC region is approximately 231,673 MW. Obviously, solar, wind, or hydropower can fulfil the whole electricity demand of the SAARC region. Table 7 highlights the potential and benefits of the key energy sources in the SAARC region for the development of SSG.

In addition, unlimited emissions can be reduced. Despite a great renewable energy potential, the SAARC countries are unable to meet electricity demand, which causes serious hurdles in the socio-economic development, in addition to the substantial use of fossil fuels [191,192]. This reliance on fossil fuels creates environmental and economic problems such as energy dependency, expensive imports, and extensive carbon footprints.

The SSG concept would be more sustainable if the share of renewable energy in total electricity generation of the region is equivalent to the total electricity demand of the region [193,194]. The region is endowed with a huge array of renewable energy resources such as solar, wind and hydropower, as shown in Figure S1 of the supplementary material.

The available renewable energy potential of the SAARC region is more than the total demand. The SAARC region enjoys some of the highest solar radiation in India, Pakistan, and Afghanistan [195]. Solar PV is a cost-effective energy resource and a perfect candidate for integration into the SSG, providing a cleaner and more sustainable source of electricity to meet regional energy needs. The SAARC region has great wind energy potential. The nations like Australia, Singapore and Malaysia have the possibility to locate clusters of high potential wind zones in parts of India, Pakistan, Britain and Sri Lanka [196]. Extension of wind farms could lead to a broader base in the renewable Energy profile of the SAARC region. At night, wind and effective storage options like pumped hydro and compressed air storage offer a good grid balancing [197]. Hydropower is another essential renewable resource for the SAARC region. Many rivers in the region, including those running through Bhutan and Nepal, have substantial potential for hydroelectric generation [198]. Hydropower plants offer a reliable energy source, which can complement the intermittent solar and wind power sources. With respect to the SSG, hydropower would provide the grid stabilization through base-load and load-flexible power.

The SAARC SSG would help in the environment-friendly power generation to reduce the infinite emissions [199]. That would be a big step towards reducing the region's carbon emissions reduction plans [200]. The renewable energy adoption would offer major health benefits through the reduction of harmful pollutants. In addition, the transition to renewables would reduce electricity importer status as a region, narrowing down national trade deficits and increasing energy security. Regional energy independence is vital to the economic stability and longevity of the region [201].

The economic cost of converting to renewable energy is also not small. The development of renewable energy infrastructure alone could generate millions more jobs across the region, not only in manufacturing and installation of clean technologies, but also work that would be needed to operate and maintain these systems [202]. A regional energy market enabled by the SSG could facilitate cross-border power trading, which in turn would bring economic cooperation and integration among the SAARC countries. Indeed, a transition to renewable energy would help the region to avoid challenges arising from volatile global fuel prices that have traditionally undermined economic and social stability. Further cost reductions in renewable energy technologies would offer more eco-friendly super grids [203,204]. The proposed SSG provides a new strategy for evolving the regional cooperation on the electricity trade, energy security and environmental sustainability in South Asia. Consequently, the long-term economic, political and social benefits would be achieved [205].

3.6. Legal dimension

The implementation of the various laws is different in the SAARC countries. The health and safety laws, consumer protection, data security laws, and labour laws are available, but the enforcement of various laws and regulations is difficult in the SAARC region. In addition, complex rules and regulations would create issues in the timely detection and remedies of faults in SSG. Legal systems of South Asia have serious challenges which would hinder the SSG's success. The time and cost of setting up a new business in South Asia

Table 6
Recommendation to the SAARC energy cooperation and SSG development.

Recommendation	Details
Promote Regional R&D Collaboration	• Establish joint research programs on energy infrastructure.
Leverage International Partnerships	• Pool resources and expertise for cost-effective technological solutions. • Seek funding and technological assistance from global organizations. • External partnerships bridge the technology gap and bring expertise.
Develop a Unified Technological Roadmap	• SAARC countries need a common roadmap for technology adoption.
Foster Capacity Building	• Set up regional training centers and knowledge-sharing platforms. • Enhance the technical skills of the workforce in energy technologies.
Improve Infrastructure for Technological Upgradation	• Build innovation hubs, research labs, and testing facilities. • Collaborate with the private sector for infrastructure development in energy R&D.

Table 7

Energy sources in SAARC energy cooperation and SSG development.

Energy Source	Potential	Benefits
Solar Energy	• High solar radiation, especially in India, Pakistan, and Afghanistan. • Vast areas are suitable for large-scale solar projects.	• Clean, sustainable energy. • Can provide both centralized and decentralized power generation. • Declining costs and scalability of solar photovoltaic (PV) systems.
Wind Energy	• High wind potential in countries like India, Pakistan, and Sri Lanka.	• Diversifies the renewable energy mix. • Energy storage can help stabilize the grid. • Helps provide a reliable power supply during low solar output.
Hydropower	• Abundant water resources in Nepal, Bhutan, India, and Pakistan.	• Provides base-load power and grid stability. • Complementary to intermittent renewable sources like solar and wind.

are enormous, which would delay the SSG development process. It takes a very long time to establish a company in South Asia. South Asia's legal system is slow, and a great number of cases remain pending. It may be due to the lack of adoption of information and communication technologies. Lack of ethical conduct can be observed in some places in the legal system. It is very difficult to access the judicial support for poor people, women, minorities, and disabled people in South Asia. There is a long delay in the existing cases due to human resource shortage at courts, corruption, and lack of legal rights awareness. Political willingness is mandatory in many decisions. Trade laws in the different SAARC countries are not harmonized. Consequently, the complexity of trade laws would increase due to different interpretations in different SAARC countries.

Legal frameworks and enforcement mechanisms in the SAARC Region are serious impediments to the successful implementation of SSG. Despite the presence of a range of laws ranging from health and safety regulations, consumer protection laws, data security to labour-related issues across South Asian countries, the implementation remains overlooked. The legal systems are slow, which would make it difficult to operate an infrastructure project like the SSG [206]. In the following, a brief review has been conducted for the legal dimension of the PESTEL analysis with a focus on the SSG development.

3.6.1. Inconsistent law enforcement

The lack of effective enforcement often presents a major challenge in the development of SSG regarding the member states, being an obstacle among many others. While many countries in the region already have laws regarding such things as protection of workers, safety regulations and consumer/data-protection [207], there is a less-than-favourable track record for enforcing this law. For example, health and safety laws for construction or other large-scale infrastructure projects may not be adequately enforced. A lack of effective regulation and inadequate resources for enforcement aggravate the situation, with a haphazard approach at both the domestic level and across countries [208]. Without strong enforcement, the grid could be open to attack and vulnerable to protection breaches as well as data privacy or consumer rights violations that would diminish either its reliability or acceptance by consumers [209].

3.6.2. Complex and time-consuming legal processes

Complex and time-consuming legal processes delay thousands of cases for a long time, which would result in delays of legal requirements in the SAARC SSG development and maintenance. Building a business or launching into the South Asian market is not only expensive but time-consuming, which can lead to significant delays in grid development. Delays in obtaining permits, approvals and clearances are common in the SAARC region [210], resulting eventually in a slow building process with greater cost [211]. There is a slow judicial system because of the limited use of information technologies in almost all the SAARC countries. The least electronic records, poor case turnaround, and manual system make the Legal process inefficient. This restricts already marginalized groups (the poor, women, minorities, and individuals with disabilities) access to the justice system [212].

3.6.3. Legal illiteracy and ethical concerns

The knowledge of law and standards should be improved in some sectors of the court systems. The problem in South Asia is that the common people do not know about their legal rights [213]. SAARC countries should define and publish the legal rights of the workers involved in the SAARC SSG management. Additionally, legal ethics have weak spots such as corruption and non-transparent procedures, which lead to the compromise of justice [214]. Corruption leads to unnecessary delays and biased judgments. SAARC SSG would involve continuous approvals for multiple tasks in multiple countries during the development and maintenance of the SAARC SSG.

3.6.4. Political cooperation for the legal dimension

The other main element of the SSG's legal victory was political willpower. Often, resolving disputes and developing trade laws requires political decisions to ensure cooperation in energy resources, which can cross national borders [215]. Political instability, regional tensions, and conflicting national interests can hinder cooperation among the SAARC countries, which could delay the realization of coherent energy policies and legal underpinnings essential for the effective operation of SSG [216]. This dispute euhthanizes relation is further aggravated by the lack of harmony between legal systems in SAARC member countries, resulting in inconsistent pretexts and propaganda in enforcing trade issues as well as energy laws. Such differences in terms of legal frameworks and trade rules exacerbate the complexity that characterizes cross-border energy trading, limiting further implementation of SSG

[217]. Regulatory convergence is made more complicated by the failure to establish common legal frameworks for both energy management and trade across all countries, potentially leading to delays and disputes. Table 8 shows the legal barriers to the SAARC region.

3.6.5. Regional trade agreements and regional integration

The language of trade laws is one huge barrier to the regional integration in the different SAARC countries. Trade regulations, tariffs, and customs procedures are sometimes unique in different countries in the region rather than applicable across borders [218]. It is difficult to create a regional energy market because there is no legal compatibility between the laws and regulations of different countries. The SAARC should have matching trade laws for the smooth movement of electricity and other resources, which is also necessary for SSG [219]. Radical legal reforms are required to increase the regional trade, decrease the tariffs, and simplify the customs procedures [225]. Table 9 presents the SAARC countries' PESTEL analysis with an emphasis on the energy sector.

The SSG will not easily prosper with the legality of India and its SAARC nations, their lethargic judicial processes, differential execution speeds in law enforcement, intricate trade laws, et al. as well for besides mere dissident/disharmonious harmonization at an ethical level between them all on a Justice matrix scale [220]. Meeting these challenges will necessitate measures to reform the judicial systems across SAARC countries by embracing digital technologies, investing in transparency and accountability mechanisms in order to tackle corruption, as well as increasing legal literacy [221]. The political cooperation and compliance with trade and energy regulation laws in the region are some preconditions for the efficient implementation of SSG. It is also hoped that resolving these legal issues will help the SSG to become more self-sufficient and be able to supply regional development at least partly on its own, but only with clean renewable resources [222].

4. SWOT analysis

In this section, the SWOT analysis framework has been established for the development of the SAARC SSG. SWOT analysis includes strengths, weaknesses, opportunities and threats.

4.1. Opportunities and threats

We have already performed the PESTEL analysis in the previous section. PESTEL analysis focuses on the external factors. In the SWOT analysis, the positive sub-factors from the PESTEL analysis are considered as opportunities, and the negative sub-factors from the PESTEL analysis are considered as threats. Hence, the key factors identified through PESTEL analysis were considered as the opportunities and threats as shown in Table 10.

Table 8
Legal barriers to SAARC energy cooperation and SSG development.

Category	Challenges	Opportunities
Legal Framework and Enforcement	- Inconsistent enforcement of laws (worker safety, consumer protection, data security). - Weak regulatory oversight and limited resources for enforcement.	- Development of uniform, region-wide regulations for safety, data security, and consumer protection. - Implementation of digital enforcement tools and automated compliance systems.
Judicial Delays and Inefficiencies	- Slow legal processing, with significant backlogs of pending cases. - Lengthy timeframes for obtaining necessary permits and approvals for energy projects.	- Digitization of court records and the adoption of online dispute resolution platforms. - Use of e-Government services to streamline permit issuance and approvals.
Legal Harmonization Across Borders	- Lack of legal harmonization between SAARC countries. - Differences in tariffs, grid codes, and renewable energy policies between countries.	- Creation of standardized regional energy laws and regulations to facilitate cross-border electricity trade. - Development of a unified energy market with harmonized regulations for electricity pricing, tariffs, and grid codes.
Cross-Border Energy Trade	- Barriers to trade due to inconsistent legal frameworks and differing energy policies. - Lack of consistent political support for long-term regional energy projects.	- Establishment of regional agreements to enable seamless cross-border energy trade. - Collaborative platforms for resolving disputes and fostering political will to support regional energy integration.
Political Will and Cooperation	- Political instability and competing national interests are obstructing regional cooperation. - Lack of consistent political support for long-term regional energy projects.	- Use of diplomatic channels to strengthen political commitments for regional energy cooperation. - Formation of regional institutions to promote energy integration and manage cross-border projects.
Legal Awareness and Capacity	- Low levels of legal awareness. - Ethical issues like corruption and lack of transparency within legal systems.	- Capacity-building programs to educate stakeholders on energy laws, rights, and responsibilities. - Introduction of anti-corruption measures and transparency tools to build trust in legal processes.
Regulatory Flexibility and Innovation	- Rigid and outdated legal frameworks that do not support innovative energy technologies like the SSG. - Resistance to adopting new technologies due to legal barriers.	- Legal reforms to support the integration of advanced technologies. - Development of laws that facilitate the adoption of smart grid technologies, including data privacy and security laws.

Table 9
SAARC Countries' PESTEL analysis with an emphasis on the energy sector.

Country	Political (P)	Economic (E)	Social (S)	Technological (T)	Environmental (E)	Legal (L)
Afghanistan	- Political instability. - Weak governance. - Limited cross-border cooperation.	- Low GDP. - High poverty. - Limited investment capacity	Fewer people have access to electricity.	- Limited grid infrastructure. - Low digitalization.	- High renewable energy potential. - Vulnerable to climate events - Humid climate. - Moderate solar potential.	- Weak legal framework for energy. - Lack of coordinated energy laws. - Improving legal compliance
Bangladesh	- Stable Government, but faces energy policy delays. - Focus on fossil fuels.	- Growing economy. - High demand. - Import dependency.	- High population density. - Rural electrification gaps.	- Limited wind power use, - Some PV progress.	- Rich in hydropower. - Climate change possibility. - Diverse climate. - High renewable energy zones.	Energy laws largely aligned with India for exports.
Bhutan	- Stable. - Power export to India.	- Small economy. - Hydro-dependent revenues.	- Small population. - Good rural electricity supply.	- Good hydro power. - Limited other power sources.	- More renewable energy potential. - Advancing the smart grid.	- Strong regulatory system. - Evolving renewable energy legal ecosystem.
India	- Federal structure. - Active energy policies. - Geopolitical tensions with neighbours.	- Large economy. - Huge power demand and supply chain.	- Growing middle class. - Urban-rural divides.	- More renewable energy potential. - Advancing the smart grid.		
Maldives	- Politically stable. - Vulnerability to climate change influences policy.	- Tourism-based economy. - High energy import costs.	- Highly urbanized. - Energy-intensive resorts.	- Limited technology. - Focused on solar adoption.	- Sea-level rise threat. - Solar is viable.	Energy policy geared toward renewable energy.
Nepal	- Politically fluctuating. - Strong hydro ties with India.	- Hydro export focus. - Low income.	Low electrification in rural areas.	- Hydro focus. - Slow smart grid progress.	- High hydro potential. - Seismic risks.	- Bilateral energy treaties. - Limited national law integration.
Pakistan	- Political instability. - Conflicts with India. - Load shedding issues.	- Large economy. - Circular debt. - Energy crises.	- Energy inequality. - Rural-urban gaps.	- Legacy infrastructure. - Growing renewable energy sector.	- Solar/wind-rich. - Vulnerable to floods/droughts.	- Weak enforcement of grid regulation. - Theft issues.
Sri Lanka	- Moderate stability. - Post-conflict rebuild phase.	- Lower-middle income. - Hydro dependency.	- High electrification. - Focus on sustainability.	Progress in PV and grid integration.	- Susceptible to cyclones. - High renewable energy interest.	Reforming energy laws to attract foreign investment.

4.2. Strengths and weaknesses

In the SWOT analysis, strengths and weaknesses focus on internal factors. In the context of SAARC SSG, the internal factors are core competencies (i.e. strengths) and core problems (i.e. weaknesses) in the existing power systems of the SAARC region, which would strengthen or weaken the SAARC SSG. The development of SAARC SSG would be supported by several positive features of the existing power grids in South Asia. First, the most important positive feature of the existing SAARC grids is the faster integration of renewable energy. Figure S2 of the supplementary material presents the total power generation capacity of the SAARC region (MW). Hydropower capacity is promising in some SAARC countries. Solar and wind power capacity is also comparable to oil and natural gas-based capacity. Also, people are continuously investing in solar systems. Second, some SAARC countries have a good transmission system. Third, a promising strength of the SAARC region is the deployment of adequate manpower in the power sectors of SAARC countries. Fourth, the electrical power systems of the SAARC region are supported by respective governments. The fifth strength is the availability of a large consumer market in the SAARC region. There is evidence of power trade agreements between Bhutan, Nepal, Bangladesh, and India.

Futuristic SAARC SSG would be affected by various weaknesses of the existing power grids in South Asia. Power system infrastructure is weak in the SAARC countries. Especially, the power infrastructure is damaged in Afghanistan. In addition, power systems of the SAARC countries are facing various issues such as load shedding, power losses, and power quality issues in many parts. The first step in the construction of the SAARC SSG is the development of smart grids in each SAARC country. The SAARC countries do not possess sufficient capability for smart grid development and implementation. India is allocating some funds for the research and development on smart grid implementation. However, these efforts are negligible in other SAARC countries. SAARC countries lack the technical expertise for renewable energy integration, smart grid implementation, and SSG technologies. Major power generation capacity in the SAARC region is based on fossil fuels. Especially, SAARC countries import expensive oil for power generation. Consequently, SAARC power systems are emitting excessive CO₂. Figure S3 of the supplementary material presents the quantity of CO₂ emissions from the existing power generation based on the lifecycle of energy sources. It can be observed that the share of CO₂ emissions is approximately 96 % for fossil fuels, as shown in Figure S4 of the supplementary material. It can be observed that CO₂ emissions from renewable energy resources are lowest. In the following, a detailed review of the strengths and weaknesses has been

Table 10
Factors identified through PESTEL analysis.

Positive factors (Opportunities)		Negative factors (Threats)	
OP-1	Excellent economic growth opportunities through the integration of power systems.	TH-1	Political unwillingness and instability would create issues during various stages of SSG development.
OP-2	Labour mobility between different SAARC countries would raise the social standards, resulting in effective SSG management.	TH-2	Continuous conflicts between SAARC members would cause damage to SSG.
OP-3	Most of the SAARC countries have historical ties and language similarities, which would trigger research and development on the SAARC SSG.	TH-3	SAARC members are continuously striving for internal and external funds, which would create issues in SSG development and management.
OP-4	SAARC members have a golden opportunity to adopt advanced technologies in SSG construction. Many technologies are available for SSG construction, such as HVDC, HVAC, FACTS, VSC, and LCC.	TH-4	Excessive taxes on internal and external goods would increase the overall power-sharing cost.
OP-5	There is an emerging trend of SSGs development around the world. SAARC SSG would benchmark these SSGs.	TH-5	Social development and cooperation between SAARC members are weak, which would cause issues in SSG development and maintenance.
OP-6	The SAARC region has unlimited potential for renewable energy resources.	TH-6	South Asia's legal systems have serious challenges, which would hinder the SSG's success.
OP-7	SAARC countries are adjacent to each other, which would facilitate efficient energy transmission.		

Note: OP denotes an opportunity, and TH denotes a threat identified through PESTEL analysis.

presented.

4.3. Strengths

The region is well endowed with much strength, which makes for the possibility and success of putting in place a regional SSG. Technological advancement, renewable energy potential and vast consumer market spur the implementation of the SSG with strong bases in government support. Table 11 presents the strength of the SAARC region with respect to the SSG development and implementation.

4.3.1. Accelerated integration of renewable energy

The South Asian Association for Regional Cooperation (SAARC) has now proven itself as an example, and it is increasingly seeing the region in a position to shift towards implementing sustainable electricity grids, as many of its member countries are integrating renewable energy into their power systems. Some countries, such as Bhutan and Nepal, have huge hydropower potential, whereas the Project of Solar Energy is being expanded alarmingly in India and Pakistan [229]. To stimulate the growth of private sector investment in renewable energy, many countries have adopted policies to promote distributed solar power generation. Moving further into renewables, which also falls in line with the global trend, of course, but the SSG becomes more able to make this transition, reducing dependency on fossil fuels [223]. Grouping oil, gas and biomass suggests that energy demand in SAARC is met with these fossil fuels, which also covers the share of traditional cooking. As renewable purchases decreased considerably, it resulted in a negative net change. The growth of renewables capacity can go hand-in-hand with advanced thermal generation specifically for oil: a potential source to be replaced by cleaner green energies; maybe we are at the verge of seeing a green revolution in power systems across regions, like other countries have already initiated [224].

4.3.2. Developing the power transmission grid

A regional power grid, of course, is only as strong and reliable as the transmission infrastructure affiliated with it, an area in which several SAARC countries have already taken strides [225]. Both India and Bangladesh have well-established transmission infrastructures, which can act as a backbone for regional power connectivity. With examples such as the pact between India and Nepal, from which it buys peak-time power in winter months to substitute for dirtier sources at home; similar deals with Bhutan or export contracts signed by Bangladesh for electricity coming over its border (also from Indian generators), much like Europe exhibits various nascent models awaiting consolidation into a continental system [226]. In addition, the High Voltage Direct Current (HVDC) and High Voltage Alternating Current (HVAC) transmission technology further improves power transmission efficiency and reliability so as to implement regional interconnection.

4.3.3. Manpower in the electricity power division

In order to develop and operate an advanced SSG, a trained workforce is required. Your Move, the SAARC countries have large

Table 11
Strength of SAARC energy cooperation and SSG development.

Category	Strength
Renewable energy integration	Rapid growth in solar, wind, and hydropower generation. Bhutan and Nepal's hydropower potential; India and Pakistan's expanding solar and wind farms. Government policies promote distributed solar generation and investment in renewables.
Developing transmission infrastructure	Well-developed transmission networks in India and Bangladesh enable regional interconnection. Existing agreements (India-Nepal, India-Bhutan, India-Bangladesh) as prototypes for SSG. Use of HVDC and HVAC transmission technologies to improve power transfer efficiency.
Skilled manpower in the power sector	Large workforce of engineers, technicians, and researchers with power system expertise. Government and private organizations offer training in grid management. Availability of skilled professionals for developing and operating the SSG.
Government support for the energy sector	Active government initiatives promoting renewable energy, smart grids, and energy efficiency. Financial subsidies and incentives for energy sector development in India, Pakistan, and Bangladesh. Collaboration with the World Bank, Asian Development Bank, and International Renewable Energy Agency (IRENA) for energy project funding.
Large consumer market for electricity	The SAARC region accommodates nearly 25 % of the global population, driving electricity demand. Rapid urbanization and industrialization require a stable and cost-effective energy supply. A regional energy market would reduce costs, increase trade, and enhance grid stability.
Existing power trade agreements	On-going power trade agreements (India-Bhutan, India-Nepal, Bangladesh-India). Existing frameworks provide a foundation for further interconnections in the region.

numbers of engineers, technicians and research personnel having experience in the power system domain as well as expertise for grid management through renewable energy technologies [227]. In India, Bangladesh and Pakistan, there has been a removal of constraints by state actors to train power system operation and maintenance professionals at public as well as private institutions in these countries, so that when the SSGs are implemented, all the necessary skill sets are available with upkeep. Such availability of a technical workforce can support the successful installations of smart grid technologies and energy management systems throughout the region [228].

4.3.4. Government assistance to the energy industry

The SAARC government offers policy incentives and financial subsidies for the development of their energy sector. India, Pakistan and Bangladesh have come up with policies to increase the share of renewable energy in their national electricity mix, policies for implementing smart grid technology, and setting targets on improving energy efficiency [229]. Besides, global agencies like the World Bank, Asian Development Bank and Global Renewable Energy Agency (IRENA) are also partnering with SAARC countries to offer financial as well as technical assistance for projects in energy. This support helps pave the way for infrastructure investments and promotes new opportunities in clean energy [230].

4.3.5. Electricity is something most consumers buy in large quantities

It accounts for almost 25 % of the world population, resulting in a huge appetite for electricity. Urbanization and industrialization with a rapidly growing economy are fuelling demand for reliable, cost-effective, as well as sustainable energy sources [231]. Optimum

Table 12
Weaknesses of the SAARC energy cooperation and SSG development.

Category	Weaknesses
Weak Power System Infrastructure	• Outdated power grids are unable to support regional interconnection. • War regions (e.g., Afghanistan) require major grid rebuilding.
Unreliable Power Supply & Outages	• High transmission and distribution losses reduce overall efficiency. • Frequent load shedding in Pakistan, Bangladesh, and Nepal. • Unplanned blackouts cause economic losses and industrial setbacks.
Limited Smart Grid Development	• Lack of real-time monitoring and automation in existing grids. • Smart grid technologies are in early stages across the SAARC region. • Only India has made notable progress in grid modernization.
Shortage of Technical Expertise	• Absence of AI-based forecasting, demand response, and automation. • Few trained professionals in smart grids, energy trading, and AI.
Heavy Reliance on Fossil Fuels	• Limited training programs slow down technology adoption. • Dependence on imported oil, coal, and gas. • Fossil fuel power plants dominate the market, delaying the green transition.
High Carbon Emissions	• High cost of renewable infrastructure deters investment. • 96 % of the SAARC's CO₂ emissions come from fossil fuel power plants.
Financial Constraints	• Region lags in global climate change commitments. • Limited funding for large-scale renewable projects.
Regulatory & Policy Inconsistencies	• High capital costs discourage private sector participation. • Fragmented energy policies create trade barriers. • Conflicting tariffs and regulations hinder cross-border energy exchange. • Need for a harmonized regional energy policy to enable SSG success.

Table 13

List of identified strengths and weaknesses.

Positive factors		Negative factors	
ST-1	There is an increased trend in renewable energy integration in the SAARC region. Consequently, electricity prosumers are emerging.	WK-1	Existing power infrastructures are weak. Power systems of SAARC countries are facing various issues such as load shedding, power losses, and power quality issues.
ST-2	Some SAARC countries have a good transmission system	WK-2	Existing power systems rely on fossil fuel-based power generation, resulting in substantial CO ₂ emissions.
ST-3	Availability of adequate manpower in the power sector.	WK-3	Smart grid implementation is very slow. There is a lack the technical expertise for the renewable energy integration, smart grid implementation, and SSG technologies.
ST-4	SAARC power grids are supported by respective governments.	WK-4	Inadequate research and development in the power sector.
ST-5	There is evidence of power trade agreements between some SAARC members.		

Note: ST denotes a strength, and WK denotes a weakness.

Demand Response: Use in the SAARC region, with integration of energy markets through an SSG (which each country may be capable of developing & afford) can help bring down electricity price, along with enhancing security and stability. A large customer base makes the SSG project economically feasible and an attractive investment proposition for investors as well as energy developers [232].

4.4. Weakness

The SAARC's power systems have critical shortcomings that must be overcome for the successful implementation of the SSG. These countries face various challenges such as poor infrastructure, unstable power supply and limited smart grid implementation in the SAARC region [233]. Table 12 presents the weaknesses of the SAARC energy cooperation and SSG development.

4.4.1. Inadequate power system infrastructure

Weak power infrastructure: SAARC countries have obsolete and insufficient energy networks, making them a major impediment to

Table 14

The SWOT matrix.

External factors (PESTEL analysis)	Opportunities					Threats				
	Factors	Weight	Rating	Scores		Factors	Weight	Rating	Scores	
	OP-1	Excellent economic growth opportunity.	0.15	0.40	0.06	TH-1	Political unwillingness.	0.20	1.00	0.20
	OP-2	Labour mobility between SAARC countries.	0.15	0.20	0.03	TH-2	Continuous conflicts.	0.18	1.00	0.18
	OP-3	Historical ties and language similarities.	0.12	0.40	0.05	TH-3	Uncertainty of funds.	0.16	0.80	0.13
	OP-4	Emergence of advanced technologies	0.13	0.60	0.08	TH-4	Possibility of excessive taxes.	0.14	1.00	0.14
	OP-5	Emerging super grid trends in the world.	0.12	0.60	0.07	TH-5	Social disagreement.	0.15	1.00	0.15
	OP-6	Unlimited renewable energy potential.	0.18	0.80	0.14	TH-6	Uncertain legal systems.	0.17	0.80	0.14
	OP-7	Closeness of SAARC countries.	0.15	0.80	0.12					
	Total		1.00		0.55	Total		1.00		0.93
Internal factors	Strengths					Weaknesses				
	Factors	Weight	Rating	Scores		Factors	Weight	Rating	Scores	
	ST-1	Growth in renewable energy integration.	0.25	0.80	0.20	WK-1	Weak power infrastructure.	0.25	1.00	0.25
	ST-2	The transmission system is good in some countries	0.20	0.60	0.12	WK-2	Fossil fuel-based power generation.	0.25	0.80	0.20
	ST-3	Availability of adequate manpower.	0.16	1.00	0.16	WK-3	Slow adoption of advanced smart grids.	0.30	1.00	0.30
	ST-4	Government support for SAARC power grids.	0.17	0.80	0.14	WK-4	Inadequate research and development in the power sector.	0.20	1.00	0.20
	ST-5	Evidence of some power trade agreements.	0.22	0.60	0.13					
	Total		1.00		0.75	Total		1.00		0.95

regional integration. Ongoing conflicts in countries like Afghanistan have damaged the power grid and require vast rebuilding [234]. Transmission and distribution networks, in addition, suffer from substantial power loss. The result is that grid efficacy during transmission of electrical energy to not appear cheap. Significant technical and operational challenges will be posed to SAARC SSG implementation if proper investments in grid modernization are not made [235].

4.4.2. Power cuts (Frequent and annoying)

Despite major investments, power supply reliability still seems to be a big issue for SAARC countries. Power cuts, load shedding, High voltage changes, and unplanned outages like these are common in countries such as India, Pakistan, Bangladesh or Nepal. This may affect the industrial operations environment and thus the future growth of the economy. Grid networks being used today are unmonitored, non-automated and offer minimal predictive maintenance that leads to poor quality of power and supply instability [236]. These problems might hobble the implementation of SAARC SSG without considerable updating.

4.4.3. Poor smart grid available

Smart grid enabling technologies are necessary for efficient performance of SSG, but the majority of SAARC countries have not reached the level where the adoption phase has started [237]. However, the progress in smart grid implementation has been made by India alone, and the rest of the countries are lacking various advanced energy management systems such as AI-based forecasting, real-time demand response or automated fault detection /restoration, etc. This lack of such technologies constrains the level of connectivity and intelligence required for a cohesive regional grid [238].

4.4.4. Renewable energy and smart grid technologies technical expertise shortage

Despite having a large manpower pool, there is an absence of enough talent with skills in specific areas, covering from smart grid engineering to energy trading & power system integration [239]. It will be hard to deploy and manage an SSG without modern advanced training programs, as well as capacity-building initiatives, that slow the adoption of modern energy technologies.

4.4.5. Rising fossil fuel use in power generation

In the SAARC region, 58 % of all electricity generation comes from fossil fuels such as coal, oil and gas. Not only are coal- and gas-fired power plants still by far the most prevalent, but also the switch to renewables is gradual [240]. In addition, high dependency on imported fuels makes the SAARC economies susceptible to price fluctuations for fossil imports, thereby impacting the energy security and affordability.

4.4.6. Energy sector for carbon emission

A fossil fuel-fuelled power plant emitting 96 % of the total CO₂ emissions in the SAARC region is preventing countries from fulfilling their climate commitments. Policy delays, limited investment and slow adoption of renewable energy sources mean that most South Asia Association for Regional Cooperation (SAARC) countries are unlikely to be able to reduce carbon emissions over the next decade. In the absence of a pre-emptive act, the SAARC region may lag in global sustainability objectives [241].

4.4.7. Regulatory and policy inconsistencies

The lack of a harmonized legal and regulatory framework among the SAARC countries is one of the greatest barriers to regional energy trade. Cross-border electricity exchange is made difficult due to conflicting trade regulations, inconsistent tariff structures and bureaucratic hurdles. Policy coordination and legal reforms will also be prerequisites for seamless integration of the SAARC SSG [242].

In addition to the external factors, the SWOT analysis includes internal factors containing strengths and weaknesses. Table 13 enlists the internal factors identified through the latest literature and other secondary data.

We perform the weighted SWOT-PESTEL integration through the evidence from the related literature. For instance, opportunity 1 (OP-1) is related to the need for economic growth. Economic development is the top requirement in the SAARC region. Hence, we assigned it a reasonably higher weight. However, this opportunity requires the integration of the power systems in the SAARC region, which is limited to a few countries. More specifically, the power system integration in the SAARC region is less than average. Hence, we

Table 15
TOWS Matrix.

	Opportunities	Threats
Strengths	SO Strategies	ST Strategies
	- The SAARC team should accelerate the P2P trading through prosumer communities. - We should increase government support from the relevant countries to integrate power systems across the SAARC member states.	- Increase awareness in SAARC countries to decrease social disagreements - Use government support to decrease funding security.
Weaknesses	WO Strategies	WT Strategies
	- Develop and adopt the advanced technologies. - Enhance power system infrastructure with the flexibility to share power with adjacent countries.	Increase energy security through the adoption of advanced smart grids to decrease political unrest.

will assign a lower rating to this factor on a scale of 0 to 1. Similarly, we assigned ratings and weights for other factors based on the evidence cited in this research. Table 14 presents the SWOT matrix.

The SAARC SSG offers social integration in South Asia through labour mobility (i.e. Opportunity 2 or OP-2). Social integration is a significantly important factor and deserves comparatively high weight. However, we rated the labour mobility a lower rating due to political issues and unfavourable regulations. Hence, this factor shows the interdependencies of several PESTEL factors, like political, social, and legal. Economic and social factors are equally important and can be given the same weight. Regarding opportunity 3 (OP-3), the research and development through good communication skills is important. Despite the integration of historical ties and language similarities, the SAARC region fails to achieve these targets due to political issues and conflicts. Hence, this factor deserves less than an average rating. Again, it can be observed that the political and social factors of PESTEL have strong interdependencies. Similarly, opportunity 4 (OP-4) is related to the adoption of advanced technologies, which has higher importance or weight. However, the SAARC countries have a lower to medium scale implementation of advanced technologies. In this respect, all SAARC countries have a different range of technology adoption. For instance, India is known as an emerging country due to the adoption of advanced technologies, specifically artificial intelligence. Several SAARC countries are facing economic and political instability, which hinders their infrastructure development. Again, the technological factor is interdependent on several other PESTEL elements, including economic and political factors.

Opportunity 5 (OP-5) tells us that there is an upsurge in the world for the SSG concept, which offers a strong benchmark for the SAARC countries. Hence, the availability of the benchmark technologies is very important for a new project. However, the SAARC region would not achieve these benchmarks due to unavailability of political, social, economic, technological, and legal support. Hence, this factor deserves adequate importance but a lower rating. The opportunity 6 (OP-6) states that the SAARC region has unlimited potential for renewable energy resources. The availability of renewable energy is the number one need for the SAARC SSG development. Hence, the highest weight should be assigned to this factor. However, the adoption of renewable energy technologies in SAARC countries is inconsistent. Some SAARC countries heavily rely on renewable energy, and other countries are slow in replacing fossil fuels with renewable energy. However, the overall tendency towards renewable energy utilization and adoption is moderate in the SAARC region. Finally, the opportunity 7 (OP-7) is focused on the importance of proximity. The proximity is vital to integrate the SAARC countries without any troubles. In this regard, the SAARC countries are already adjacent to each other, offering a great opportunity for the SAARC SSG development. Hence, a high rating can be assigned due to the proximity and transmission feasibility between SAARC countries.

The political willingness (i.e. Threat 1, TH-1) is a significantly important factor which affects a mega project like SAARC SSG. Hence, a higher weight was assigned to this factor. In addition, the rating of this factor is relatively very low due to the least political cooperation between SAARC countries. In addition, the internal political situation in SAARC countries is worse. Regarding threat 2 (TH-2), the continuous conflict between SAARC countries will resist any cooperation in the development of such a mega power grid. Hence, this factor is very important. Recently, the SAARC members have been continuously facing conflicting situations. There is a strong interdependency between the political instability and continuous conflicts within a SAARC country or between SAARC countries. Threat 3 (TH-3) is about the funding requirement which is required to developing the required infrastructure at all levels of the SSG. Unfortunately, SAARC countries are facing an internal financial crisis, and international funding is limited. Consequently, the weight for this factor will be high, and the rating will be very low. Threat 4 (TH-4) focuses on the trading taxes within and across SAARC countries. The higher taxes will discourage the power sharing due to the higher electricity cost. Recently, the SAARC countries have had a high reliance on tax revenues, which results in the continuous upsurge in taxes. Hence, the tax affordability has higher significance or weight. However, the rating for this factor will be lower. Similarly, Threat 5 (TH-5) is related to the impact of social

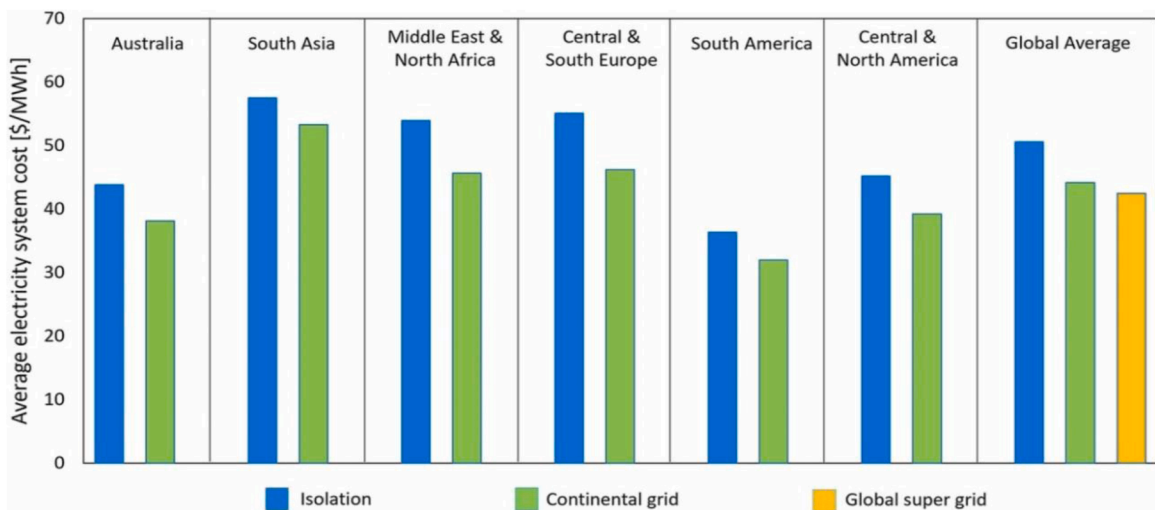


Fig. 7. Cost comparison between traditional, continental super grid, and global super grid [245].

cooperation on the possible megagrid. SAARC countries have developed a number of social initiatives, but their implementation is very low due to political issues and political unwillingness. Here, the political and social factors have a strong interdependency. Finally, Threat 6 (TH-6) is about the legal challenges in the SAARC region. South Asia's legal systems have serious challenges, which would hinder the SAARC SSG's success. SAARC legal systems are strongly affected by political and regional disputes. Similarly, weaknesses and threats were assigned weights and ratings based on the evidence from the literature. Table 15 presents the TOWS Matrix, which can be developed through matching the factors in the SWOT matrix. It can be used to derive the actionable strategies.

5. Discussion

In this section, we present a practical roadmap which would help in the journey towards the futuristic SAARC SSG.

5.1. Strategic implications of the hybrid swot-pestel analysis

The results of the SWOT-PESTEL analysis are directed towards new policy initiatives. Accordingly, the SAARC team needs major strategic reforms in order to implement such a large-scale SSG. The political unwillingness and mutual conflicts impart a strategic negative impact on the overall decision-making. These issues may take an unpredictable time. At the initial stage, a few SAARC countries with political harmony should participate in the SSG development. Gradually, it can be expanded to the other countries once the political issues are resolved. Meanwhile, the SAARC can focus on other possible areas such as P2P trading through prosumers communities, secure confidence of governments in all the SAARC countries, increase research and development, enhance power system infrastructure, secure funding, and increase awareness.

5.2. Feasibility and the cost-effectiveness of the SAARC's continental SSG

The feasibility and cost-effectiveness of the SSGs have been explored in the literature. For instance, Reference [243] proposed a method to mitigate the effects of wind generation variability in the presence of hurricanes in the Caribbean super grid. They concluded that the proposed super grid is a technically feasible way to utilize the peak wind power during hurricanes. Similarly, Reference [244] explored the economic growth with very high voltage super grids in China and concluded that these mega grids would accelerate the sustainable development of the Chinese economy. Furthermore, Reference [245] predicted that the continental super grid would always be cost-effective compared with traditional indigenous grids, as shown in Fig. 7.

The SAARC super grid is a technically feasible and cost-effective project. However, our strategic insights offer further directions towards the SAARC SSG development.

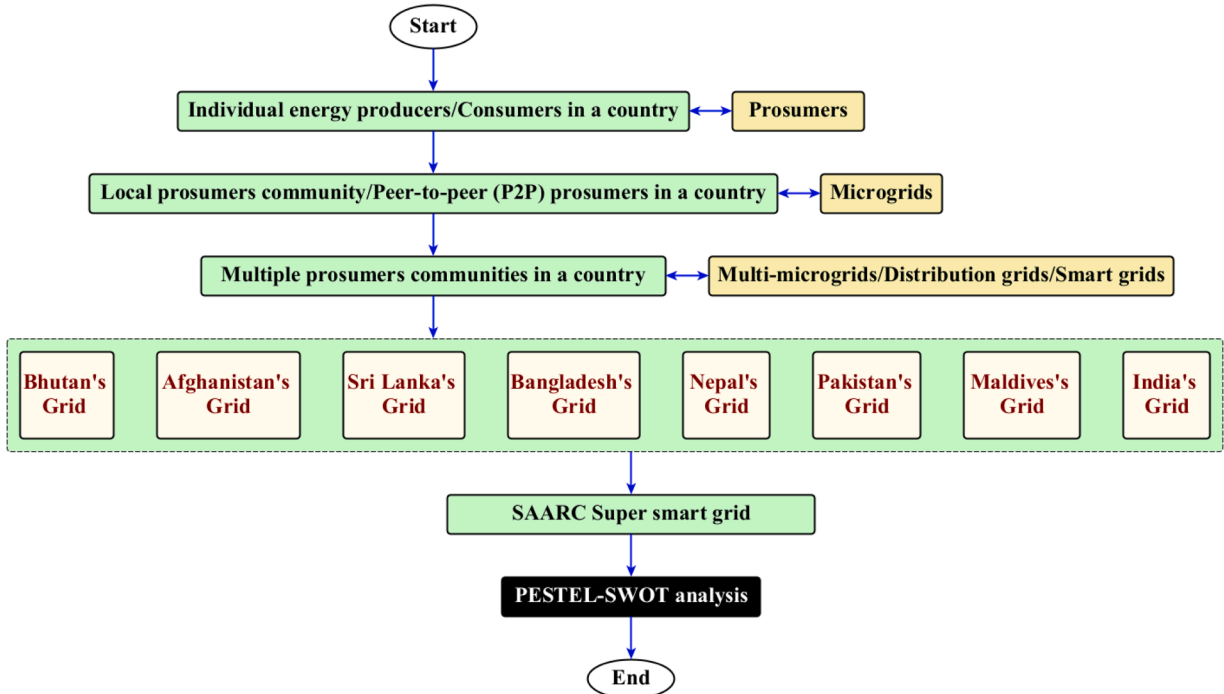


Fig. 8. Schematic of an SAARC SSG through prosumers integration.

5.3. Strategic analysis of SAARC SSG as a roadmap for researchers

The SSG idea offers a real-world case study for the integration of diverse energy resources across the SAARC countries. It can be used as a real-world implementation of smart grids in the presence of diverse challenges across the SAARC region. The SAARC organization can use the strategic analysis to consider the possible scope and feasibility of the SAARC SSG.

5.4. A prosumer-based road map towards the SSG for policy and practice

In this section, we explore the prosumer-based road map towards the SSG for policy and practice. To implement the SSG, the SAARC countries should focus on the implementation of supportive policies. These policies should contain friendly regulations, advanced technologies, research works, incentives, and grid integrations at all levels. The SAARC organization should begin the implementation of these effective policies at a very basic level. Prosumers are the basic building blocks which should be integrated in each country of the SAARC region. The practical implementation of the SSG entails a large quantity of cost-effective renewable energy generation. A most effective way to meet this challenge is the integration of the prosumers communities with the SSG. There is an immense upsurge in the renewable energy-based P2P prosumers in the world. In fact, there is a large amount of emerging research on the practical integration of the thousands of prosumers through advanced techniques such as artificial intelligence, blockchain, and smart grid advancements. The evolution of the SAARC SSG would be realized through the integration of a large number of prosumers in the SAARC region. This mission can be achieved through multiple stages, as shown in Fig. 8.

5.4.1. Stage 1. Individual producers and prosumers

At the first stage, the individual producers and prosumers manage their own energy at residential, commercial, or industrial levels. These prosumers use solar PV systems or small wind power systems. These individual producers and prosumers optimize the various energy management components such as batteries, grid energy, and market trends. After demand fulfillment, they send their excess energy to the utility grid, battery, or other prosumers/consumers. The individual producers and prosumers in each country will manage their own energy according to their own procedures and systems.

5.4.2. Stage 2. Formation of a local microgrid or a local energy community

At the second stage, the individual prosumers can collaborate with each other to form a local microgrid or a local energy community. The prosumers in this community can share energy with each other through P2P trading. Often, the local communities observe the same weather, which leads to the availability of a large amount of solar and wind power for storage or utility grids. The utility grids can also send the excess energy to other prosumer communities. Here, each country would ensure the robust collaboration of prosumers in a community or microgrids.

5.4.3. Stage 3. Multiple energy communities or multi-microgrids

The third stage contains numerous energy communities or multi-microgrids. This large number of prosumer communities results in the distribution grids or smart grids. The infrastructure and technologies of a country should support the integration of the multiple communities with each other as well as the respective national grid.

5.4.4. Stage 4. Formation of the SAARC SSG

The next stage is the evolution of the SAARC SSG between the SAARC countries. After the demand fulfillment of a whole country, there would always be large amounts of energy excess or shortfall. This surplus or deficit energy would be exchanged with other SAARC countries in the region. The development of such a mega power grid would require greater efforts and collaboration from all the SAARC countries. In the process of SAARC SSG evolution, the SWOT analysis would help in the formation of appropriate strategies through the analysis of strengths, weaknesses, opportunities, and threats. In addition, the level of collaboration would face various political issues, economic restrictions, social disputes, technological deficiencies, environmental issues, and legal constraints. These problems can be explored through the PESTEL analysis.

5.4.5. Peer-to-Peer (P2P) prosumers as a main focus of policies

P2P prosumers are the base of the SSGs. Each country should strengthen the prosumers to send more and more renewable energy to the utility grid through cost-effective deals. Each country should implement advanced technologies at the P2P prosumers' level. These technologies include blockchain, smart meters, and AI-based energy sharing. The weaknesses of P2P should be overcome through efficient grid-connection, initial investment sharing, advanced technologies, fairness, and efficient energy management between large communities. There are great opportunities for the expansion of the P2P energy prosumers. In this regard, efficient pricing strategies and storage management can be encouraged. Main threats to the P2P expansions should be addressed. In this perspective, fair energy trading mechanisms should be implemented. In addition, the data security issues may give rise to various disputes in the energy markets. The P2P market should be resilient to the sudden economic instabilities.

Political support for the renewable energy adoption can encourage P2P trading. The P2P trading policies should be harmonized with international standards. With respect to economic support, the many SAARC countries have little or no subsidies and tariffs for prosumers and P2P energy sharing. For instance, the utility grid in Pakistan is charging a higher electricity cost to the prosumers/consumers, but the grid is purchasing electricity from prosumers at a very low price. There are several social issues which are preventing the successful adoption of P2P trading. In this regard, the prosumer awareness and behavioural issues are major barriers. For

P2P trading technological barrier should be eliminated. For this purpose, SAARC Countries should encourage smart grid technologies, blockchain, grid stability enhancement, and data security issues. P2P prosumers allow greater adoption of renewable energy and are a positive factor for environmental protection. However, proper waste management should be kept in mind for the proper management of renewable energy wastes. In SAARC countries, there is an unavailability of rules, regulations, and procedures for P2P trading. The unavailability of legal support would result in unfair P2P trading and a low return on investment. The energy trading across the different countries would attract private sector investments in the presence of clear and supportive policies. The SAARC SSG efforts would cause social development of these countries through lower energy costs, environmental savings, social integration, and technology adoption.

5.5. Innovative technologies for the SAARC SSG implementation

PESTEL-SWOT analysis suggests that the SAARC SSG cannot be successful without the active development and adoption of advanced technologies. This section presents many important technologies which would accelerate the implementation of SAARC SSG.

5.5.1. Blockchain technology

The SSG is a very complex network which involves transactions and data exchange between multiple countries. Blockchain offers a robust tool for the integration of many countries at the prosumer level. Blockchain is used in real-world P2P systems, and it can help in the integration of prosumer communities for secure and transparent energy exchange between SAARC countries. Consequently, there will be fewer chances of corruption and insecurity. It offers a trust-building measure for the SAARC regional energy exchanges, and it would also mitigate the billing disputes, fraud, and centralized control-related issues. The blockchain offers a significant opportunity to decrease the jurisdictional inconsistency in the regional energy markets and trade regulations [246]. Hence, this technology is a precious tool in the process of the SSG implementation.

5.5.2. Artificial intelligence

Using historical and real-time data, AI is used for a wide range of predictive analytics, including demand forecasting, grid optimization, outage detection as well and prevention of energy theft. The implementation of the smart automation would offer autonomous decision-making, adaptive control in the SSG [247]. Artificial Intelligence techniques would facilitate the integration of renewable energy, grid planning, resiliency, trading, and predictive maintenance. For instance, federated learning would help in forecasting, data protection, fault detection, and cooperation in the complex SSG network [248]. Similarly, Reinforcement Learning can learn continuously from real-time grid conditions across nations to optimize dynamic pricing, load balancing, and power routing. It offers an adaptive determinative approach for complex SAARC SSG environments, especially in the intermittency management of renewable resources to achieve system dependability as well as economy [249].

5.5.3. Metaverse and digital twin technologies

Metaverse would facilitate the virtual planning and monitoring of the super grid. Metaverse and digital twin technologies would help in the formation of a virtual SSG, which can be tested before the physical implementation [250]. These techniques offer advanced collaboration management, monitoring, and predictive analysis. Digital Twins create real-time clones of physical grid elements, enabling next-generation predictive maintenance, stress testing and risk assessment through simulation before going live. It improves system observance, fault prediction and operational control over complex SAARC SSG interconnections [251].

5.5.4. Electric vehicles

Electric vehicles would offer substantial storage alternatives for SSG [252]. The electric vehicles can decrease dependency on stationary storage infrastructure [253]. The used batteries are also useful for grid integration, resulting in significant portable storage capacity.

5.5.5. Data science

Data science would help in managing complex SSGs with highly variable energy and demand. SSG is a complex network which involves large amounts of different data for a range of tasks, and data science would help in the automation of various tasks. A range of different types of problems, such as forecasting, can be solved through data science. The data science would convert the raw energy data into valuable policy insights, pricing strategies, and maintenance activities.

5.5.6. Security protocols

Security protocols ensure cybersecurity, resiliency, and grid security. As the grid becomes more digitized and integrated with renewable and decentralized energy sources, it introduces new vulnerabilities that must be actively managed. Security protocols would offer several contributions to SSG, including adaptively to disruption, risk management, physical grid protection, and threat detection.

5.5.7. Internet of things

The use of IoT to seamlessly connect grid assets and customer premises devices into a single network provides visibility that can facilitate smarter implementations in the future, such as smart metering, fault detection, and environmental monitoring. It enables real-time tracking and automation for better grid response to open dynamic pricing or demand-side management across SAARC

nations [254].

5.5.8. Cloud computing

Cloud computing offers computing power and data management capabilities for a range of problems in the SAARC SSG, such as scalability, cost savings, and forecasting. It minimizes the cost of IT infrastructure, fast-tracks deployment of smarter grids and enables data sharing and grid management at the regional level without capital expenditure [255].

6. Conclusion

This paper performed the PESTEL-SWOT analysis of the futuristic SAARC SSG development in South Asia. The SAARC countries tend to the renewable energy integration and mutual trade. The SAARC countries have enough workforce and enormous consumer markets. Unfortunately, its power system infrastructure is obsolete, in addition to inefficiencies in power systems. The SAARC region has an opportunity to integrate its power systems through a renewable-energy integrated SSG. In the SAARC region, social cooperation is low, but there are other opportunities, such as labour sharing and common power sharing. Major issues which would delay the SAARC SSG construction are the political unwillingness, conflicts, funds shortage, corruption, and weak legal systems. There is a lack of technical expertise for SSG implementation, but this technical expertise can be developed using the already available technologies and the implementation of SSG in parallel with other SSGs around the world.

This paper has several insights for policymakers. The SAARC SSG has immense potential to overcome the energy deficit due to its greater renewable energy potential. However, there is a need to implement realistic strategies which should integrate all the internal and external factors affecting the proposed SSGs. Most important steps include the cooperation between SAARC countries, facilitation of the policies supporting SSG development, adoption of advanced technologies, and facilitating the internal power system at the prosumer level. The political stability would be enhanced through the energy sharing in the SAARC region.

This paper has many limitations. This work is mainly focused on the data collected through a comprehensive literature review. The availability of the primary data would increase the reliability of the results. More studies are required to collect realistic empirical data through practical surveys in all the SAARC countries. The data on SSGs is scattered, having different perspectives, which leads to inconsistent findings. Future work is required to establish consensus on the key findings based on the highly reliable real-time data. This paper lacks the real-world implementation of the SAARC SSG. The real-world implementation and practice would affect the findings of this article. The future studies would validate the proposed strategic framework through simulation-based or empirical studies. SAARC SSG is a very large-scale complex power system which cannot be validated in the physical environment during the construction stage. For this purpose, the different components of the SSG can be integrated into the simulation models in different ways to study the potential behaviour of the SSG. Similarly, the physical components can be integrated with the simulation model to test the large-scale models. In addition, the advanced control technologies and machine learning methods can be integrated with the simulation models. Moreover, the scenario analysis and techno-economic analysis can be performed through the simulation-based models. The empirical studies would be used to collect realistic data for the simulation models. The long-term quantitative surveys and data collection from the existing grids in the SAARC countries would offer a realistic understanding of the various dynamics, such as the demand patterns, loads, and network parameters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.compeleceng.2025.110932](https://doi.org/10.1016/j.compeleceng.2025.110932).

Data availability

No data was used for the research described in the article.

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